36

Image formation

CHAPTER OUTLINE

36.1 Images Formed by Flat Mirrors

36.2 Images Formed by Spherical Mirrors

36.3 Images Formed by Refraction

36.4 Images Formed by Thin Lenses

36.5 Lens Abberations

36.6 The Camera

36.7 The Eye

36.8 The Simple Magnifier

36.9 The Compound Microscope

36.10 The Telescope

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ36.1 Answer (b). A change in the medium in contact with the outer surface will result in a change in refraction at the outer surface if the surface is curved. Refraction should be limited to the inner surface because the medium inside (air) does not change. The outer surface should be flat so that it will not produce a fuzzy or distorted image for the diver when the mask is used either in air or in water.

OQ36.2 **(i)** Answer (c). The image is an upright and virtual at first then inverted and real. A concave (converging) mirror can produce real and virtual images depending on the object distance.

(ii) Answer (c). When the object passes through the focal point, the image switches from virtual to real.

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OQ36.3 Answer (b). A converging lens forms real, inverted images of real objects located outside the focal point.

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}: \qquad \frac{1}{50.0\,\text{cm}}+\frac{1}{q}=\frac{1}{15.0\,\text{cm}} \quad \rightarrow \quad q = 21.4\,\text{cm}$$

$$M=\frac{-q}{p}=\frac{-21.4\text{ cm}}{50.0\text{ cm}}=-0.429$$

The positive image distance confirms that the image is real, and the negative magnification confirms that the image is inverted. Also, $M = -0.429$ tells us the image is smaller than the object.

OQ36.4 **(i)** Answer (e). A converging lens forms real, inverted images of real objects located farther than the focal length ($p > f$), and virtual, upright images of real objects located closer than the focal length ($p < f$).

(ii) Answers (a) and (c). A diverging lens forms a virtual, upright, and diminished image of any real object located any distance from the lens.

OQ36.5 Answer (d). The entire image is visible, but only at half the intensity. Each point on the object is a source of rays that travel in all directions. Thus, light from all parts of the object goes through all unblocked parts of the lens and forms an image. If you block part of the lens, you are blocking some of the rays, but the remaining ones still come from all parts of the object.

OQ36.6 Answer (d). The image is upright, so the magnification is positive:

$$M=\frac{-q}{p}: \qquad +1.50=\frac{-q}{30.0\text{ cm}} \quad \rightarrow \quad q = -45.0\text{ cm}$$

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}: \qquad \frac{1}{30.0\text{ cm}}+\frac{1}{-45.0\text{ cm}}=\frac{1}{f} \quad \rightarrow \quad f = 90.0\text{ cm}$$

OQ36.7 Answer (b). For lens 1, the object distance $p_1 = 50.0$ cm:

$$\frac{1}{p_1}+\frac{1}{q_1}=\frac{1}{f_1}: \qquad \frac{1}{50.0\text{ cm}}+\frac{1}{q_1}=\frac{1}{15.0\text{ cm}} \quad \rightarrow \quad q_1 = 21.4\text{ cm}$$

The image distance is positive, so the image is real and forms 21.4 cm to the right of lens 1.

The image of lens 1 is the object of lens 2. For lens 2, the object distance $p_2 = 35.0\text{ cm} - 21.4\text{ cm} = 13.6$ cm:

$$\frac{1}{p_2}+\frac{1}{q_2}=\frac{1}{f_2}: \qquad \frac{1}{13.6\text{ cm}}+\frac{1}{q_2}=\frac{1}{10.0\text{ cm}} \quad \rightarrow \quad q_2 = 38.0\text{ cm}$$

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The image distance is positive, so the image is real and forms 38.0 cm to the right of lens 2.

From Equation 36.18, the overall magnification is

$$M = M_1 M_2 = \left(\frac{-q_1}{p_1}\right)\left(\frac{-q_2}{p_2}\right) = \left(\frac{-21.4\,\text{cm}}{50.0\,\text{cm}}\right)\left(\frac{-38.0\,\text{cm}}{13.6\,\text{cm}}\right) = 1.20$$

OQ36.8 Answer (c). The amount of light focused on the film by a camera is proportional to the area of the aperture through which the light enters the camera. Since the area of a circular opening varies as the square of the diameter of the opening, the light reaching the film is proportional to the square of the diameter of the aperture. Thus, increasing this diameter by a factor of 3 increases the amount of light by a factor of 9.

OQ36.9 Answer (b). The angle of refraction for the light coming from the fish to the person is 60°. The angle of incidence is smaller, so the fish is deeper than it appears. [Refer to CQ35.16.]

OQ36.10 The ranking is c > e > a > d > b. In case (c) the object distance is effectively infinite. In (e) the object distance is very large compared to the focal length, but not infinite. In (a) the object distance is a little larger than the focal length. In (d) the object distance is equal to the focal length. In (b) the object distance is less than the focal length.

OQ36.11 Answer (d). We can answer this question conceptually by noting that if the lens were surrounded by water, parallel light rays passing into and out of the lens would experience smaller changes in the index of refraction, so they would bend less, and so would focus farther from the lens.

We can answer this question quantitatively if we consider the derivation of the lens makers' equation (Equation 36.15) for the general case of the lens being surrounded by a medium of index n_0. We would conclude that Equation 36.15 takes the general form

$$\frac{1}{f} = \left(\frac{n}{n_0} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

So, for a lens of crown glass (n = 1.52, from Table 35.1) surrounded by air, n_0 = 1, we have

$$\frac{1}{f} = (1.52 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{15.0\ \text{cm}}$$

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but for a lens surrounded by water, $n_0 = 1.333$, and we have

$$\frac{1}{f} = \left(\frac{1.52}{1.333} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= \frac{\left(\frac{1.52}{1.333} - 1\right)}{(1.52 - 1)}\left[(1.52 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)\right] = \frac{\left(\frac{1.52}{1.333} - 1\right)}{(1.52 - 1)}\frac{1}{15.0 \text{ cm}}$$

$$f = 55.6 \text{ cm}$$

OQ36.12 Answer (e). At the smallest distance the object and image distances are equal, $p = q$:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}: \quad \frac{1}{p} + \frac{1}{p} = \frac{1}{f}$$

$$\frac{2}{p} = \frac{1}{f} \quad \rightarrow \quad p = 2f = q$$

OQ36.13 **(i)** Answers (a) and (c). The image of a real object formed by a plane mirror is always an upright and virtual image, which is the same size as the object and located as far behind the mirror as the object is in front of the mirror.

(ii) Answer (e). A concave (converging) mirror forms real, inverted images of real objects located outside the focal point ($p > f$), and virtual, upright images of real objects located inside the focal point ($p < f$) of the mirror.

(iii) Answer (a) and (c). With a real object in front of a convex (diverging) mirror, the image is always virtual, upright, and diminished in size, and located between the mirror and the focal point.

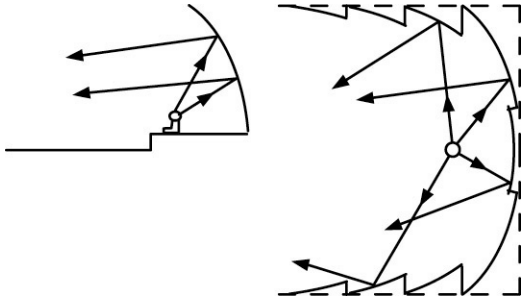
OQ36.14 Answer (b). The image is upright, and corresponding parts of the object and image are the same distance from the mirror.

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ANSWERS TO CONCEPTUAL QUESTIONS

CQ36.1 (a) Yes.

(b) You have likely seen a Fresnel mirror for sound. The diagram represents first a side view of a band shell. It is a concave mirror for sound, designed to channel sound into a beam toward the audience in front of the band shell. Sections of its surface can be kept at the right orientations as they are pushed around inside a rectangular box to form an auditorium with good diffusion of sound from stage to audience, with a floor plan suggested by the second part of the diagram.



ANS. FIG. CQ36.1

CQ36.2 (a) The focal point is defined as the location of the image formed by rays originally parallel to the axis. An object at a large but finite distance will radiate rays nearly but not exactly parallel. Infinite object distance describes the definite limiting case in which these rays become parallel.

(b) To measure the focal length of a converging lens, set it up to form an image of the farthest object you can see outside a window. The image distance will be equal to the focal length within one percent or better if the object distance is a hundred times larger or more.

CQ36.3 Because when you look at the AMBULANCE in your rear view mirror, the apparent left-right inversion clearly displays the name of the AMBULANCE behind you. Do not jam on your brakes when a MIAMI city bus is right behind you.

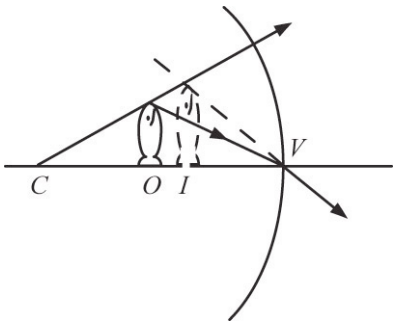
CQ36.4 Chromatic aberration arises because a material medium's refractive index can be wavelength dependent. A mirror changes the direction of light by reflection, not refraction. Light of all wavelengths follows the same path according to the law of reflection, so no chromatic aberration happens.

CQ36.5 (a) Yes. If the converging lens is immersed in a liquid with an index of refraction significantly greater than that of the lens itself, it will make light from a distant source diverge.

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(b) No. This is not the case with a converging (concave) mirror, as the law of reflection has nothing to do with the indices of refraction.

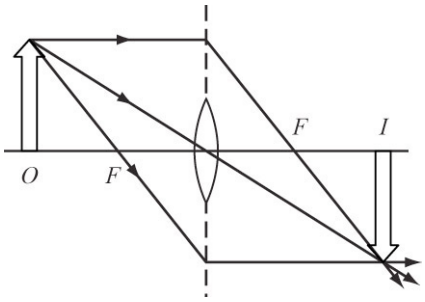
CQ36.6 As in the diagram, let the center of curvature C of the fishbowl and the bottom of the fish define the optical axis, intersecting the fishbowl at vertex V. A ray from the top of the fish that reaches the bowl surface along a radial line through C has angle of incidence zero and angle of refraction zero. This ray exits from the bowl unchanged in direction. A ray from the top of the fish to V is refracted to bend away from the normal. Its extension back inside the fishbowl determines the location of the image and the characteristics of the image. The image is upright, virtual, and enlarged.



ANS. FIG. CQ36.6

CQ36.7 (a) An infinite number. In general, an infinite number of rays leave each point of any object and travel in all directions. Note that the three principal rays that we use for imaging are just a subset of the infinite number of rays.

(b) All three principal rays can be drawn in a ray diagram, provided that we extend the plane of the lens as shown in Figure CQ36.7.



ANS. FIG. CQ36.7

CQ36.8 With the meniscus design, when you direct your gaze near the outer circumference of the lens you receive a ray that has passed through glass with more nearly parallel surfaces of entry and exit. Thus, the lens minimally distorts the direction to the object you are looking at. If you wear glasses, turn them around and look through them the wrong way to maximize this distortion.

CQ36.9 Note that an object at infinity has an image at the focal point of a converging lens, and an object at the focal point of a converging lens

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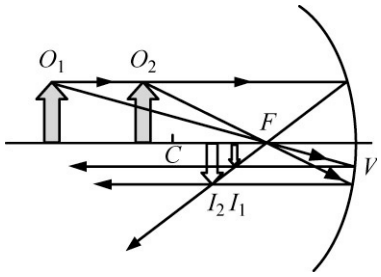
has its image at infinity, so we may conclude that the farther an object is from a lens, the closer the image is to the focal point of the lens. Therefore, we expect the image of the farther tree to form closer to the lens, so we conclude that the screen should be moved toward the lens.

We can verify our conclusion using the lens equation:

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f} \quad \rightarrow \quad \frac{1}{q}=\frac{1}{f}-\frac{1}{p}=\frac{p-f}{fp}=\frac{1-f/p}{f} \quad \rightarrow \quad q=\frac{f}{1-f/p}$$

For $p = x,\ q=\dfrac{f}{1-f/x},$ and for $p' = 2x,\ q'=\dfrac{f}{1-f/2x} < q,$ so our conclusion is correct.

CQ36.10 In the diagram, only two of the three principal rays have been used to locate images to reduce the amount of visual clutter. The upright shaded arrows are the objects, and the correspondingly numbered inverted arrows are the images. As you can see, object 2 is closer to the focal point than object 1, and image 2 is farther to the left than image 1.



ANS. FIG. CQ36.10

CQ36.11 The eyeglasses on the left are diverging lenses that correct for nearsightedness. If you look carefully at the edge of the person's face through the lens, you will see that everything viewed through these glasses is reduced in size. The eyeglasses on the right are converging lenses, which correct for farsightedness. These lenses make everything that is viewed through them look larger.

CQ36.12 The eyeglass wearer's eye is at an object distance from the lens that is quite small—the eye is on the order of 10^{-2} meter from the lens. The focal length of an eyeglass lens is several decimeters, positive or negative. Therefore the image distance will be similar in magnitude to the object distance. The onlooker sees a sharp image of the eye behind the lens. Look closely at Figure CQ36.11a and notice that the wearer's eyes seem not only to be smaller, but also positioned a bit behind the plane of his face—namely, behind where they would be if he were not wearing glasses. Similarly, in Figure CQ36.11b, his eyes

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seem to be magnified and in front of the plane of his face. We as observers take the light information coming from the object through the lens and perceive or photograph the image as if it were an object.

CQ36.13 Absolutely. Only absorbed light, not transmitted light, contributes internal energy to a transparent object. A clear lens can stay ice-cold and solid as megajoules of light energy pass through it.

CQ36.14 Make the mirror an efficient reflector (shiny). Make it reflect to the image even rays far from the axis, by giving it a parabolic shape. Most important, make it large in diameter to intercept a lot of solar power. And you get higher temperature if the image is smaller, as you get with shorter focal length; and if the furnace enclosure is an efficient absorber (black).

CQ36.15 The artist's statements are accurate, perceptive, and eloquent. The image you see is "almost one's whole surroundings," including things behind you and things farther in front of you than the globe is, but nothing eclipsed by the opaque globe or by your head. For example, we cannot see Escher's index and middle fingers or their reflections in the globe.

The point halfway between your eyes is indeed the focus in a figurative sense, but it is not an optical focus. The principal axis will always lie in a line that runs through the center of the sphere and the bridge of your nose (between your eyes). Outside the globe, you are at the center of your observable universe. If you close one eye, the center of the looking-glass world may hop over to the location of the image of your open eye (depending on which eye is dominant).

CQ36.16 Both words are inverted, but the word OXIDE looks the same when inverted.

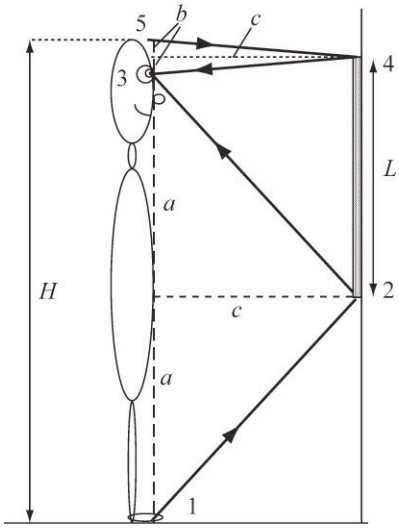
CQ36.17 Yes, the mirror equation and the magnification equation apply to plane mirrors. A curved mirror is made flat by increasing its radius of curvature without bound, so that its focal length goes to infinity. From $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}=0$ we have $\frac{1}{p}=-\frac{1}{q}$; therefore, $p = -q$. The virtual image is as far behind the mirror as the object is in front. The magnification is $M=-\frac{q}{p}=\frac{p}{p}=1$. The image is right side up and actual size.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 36.1 Images Formed by Flat Mirrors

P36.1 ANS. FIG. P36.1 shows the path of rays reflected by a mirror of minimum height: rays from the person's feet and top of his head travel along the respective paths 123 and 543 to his eyes. The rays reflect at the bottom and top of the mirror. Because of the law of reflection, the paths can be considered to form the hypotenuses of two pairs of right triangles with common base c: two large similar right triangles with height a, and two small similar right triangles with height b.

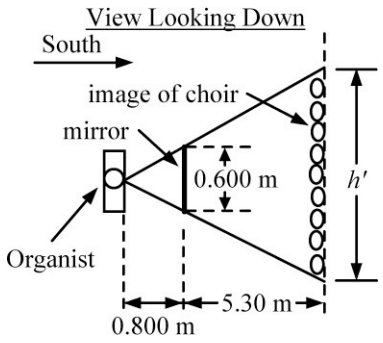


ANS. FIG. P36.1

Rays from his feet enter his eyes a vertical distance $2a$ from the ground. The rays from the top of his head enter his eyes a distance $2b$ from the top of his head. His full height is $H = 2a + 2b$. The mirror has height $L = a + b$. We see then that

$$L = a + b = \frac{H}{2} = \frac{178\text{ cm}}{2} = \boxed{89\text{ cm}}$$

P36.2 The virtual image is as far behind the mirror as the choir is in front of the mirror. Thus, the image is 5.30 m behind the mirror. The image of the choir is 0.800 m + 5.30 m = 6.10 m from the organist. Using similar triangles:



ANS. FIG. P36.2

$$\frac{h'}{0.600\text{ m}} = \frac{6.10\text{ m}}{0.800\text{ m}}$$

or
$$h' = (0.600\text{ m})\left(\frac{6.10\text{ m}}{0.800\text{ m}}\right) = \boxed{4.58\text{ m}}$$

P36.3 (a) Younger. Light takes a finite time to travel from an object to the mirror and then to the eye.

(b) I stand about 40 cm from my bathroom mirror. I scatter light, which travels to the mirror and back to me in time interval

$$\Delta t = \frac{2d}{c} = \frac{0.8\text{ m}}{3\times10^8\text{ m/s}} \sim \boxed{10^{-9}\text{ s}},$$ showing me a view of myself as I was then.

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P36.4 The mirrors are 6.00 m apart.

(1) The first image in the left mirror is 2.00 m behind the mirror, or 2.00 m + 2.00 m = $\boxed{4.00 \text{ m}}$ from the position of the person.

(2) The first image in the right mirror is located 4.00 m behind the right mirror, but this location is 4.00 m + 6.00 m = 10.0 ft from the left mirror. Thus, the second image in the left mirror is 10.00 m behind the mirror, or 10.00 m + 2.00 m = $\boxed{12.00 \text{ m}}$ from the person.

(3) The first image in the left mirror forms an image in the right mirror. This first image is 2.00 m + 6.00 m = 8.00 m from the right mirror, and, thus, an image 8.00 m behind the right mirror is formed. This image in the right mirror also forms an image in the left mirror. The distance from this image in the right mirror to the left mirror is 8.00 m + 6.00 m = 14.00 m. The third image in the left mirror is, thus, 14.00 m behind the mirror, or 14.00 m + 2.00 m = $\boxed{16.00 \text{ m}}$ from the person.

P36.5 For a plane mirror, $q = -p$. Recall from common experience that the position of an image does not shift as a viewer rotates. Thus, to a viewer looking toward a mirror that is turned by 45°, the image distance still follows this rule.

(a) The upper mirror M_1 produces a virtual, actual-sized image I_1 according to

$$M_1 = -\frac{q_1}{p_1} = +1$$

As shown in ANS. FIG. P36.5, this image is a distance p_1 above the upper mirror. It is the object for mirror M_2, at object distance

$$p_2 = p_1 + h$$

The lower mirror produces a virtual, actual-sized, right-side-up image according to

$$q_2 = -p_2 = -(p_1 + h)$$

with

$$M_2 = -\frac{q_2}{p_2} = +1 \quad \text{and} \quad M_{\text{overall}} = M_1 M_2 = 1.$$

Thus the final image is at distance $\boxed{p_1 + h, \text{ behind the lower mirror}}$.

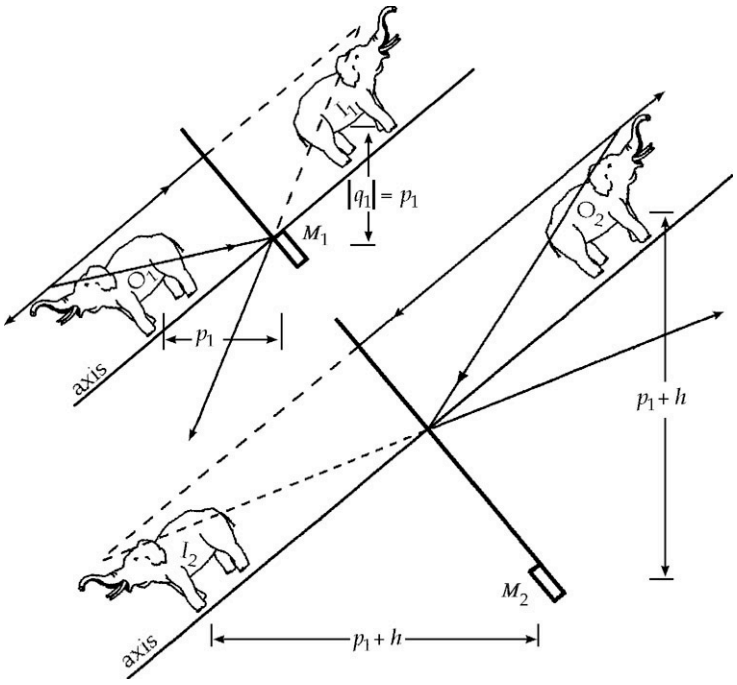
(b) It is $\boxed{\text{virtual}}$.

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(c) Upright

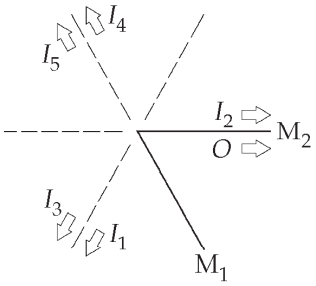
(d) With magnification +1.00 .

(e) No. Left and right are not reversed. In a top view of the periscope, parallel rays from the right and left sides of the object stay parallel and on the right and left. The first mirror switches left and right, but the second mirror switches them again; so, overall left and right are not reversed.



ANS. FIG. P36.5

***P36.6** A graphical construction, shown in ANS. FIG. P36.6, produces 5 images, with images I_1 and I_2 directly into the mirrors from the object O, and (O, I_3, I_4) and (I_2, I_1, I_5) forming the vertices of equilateral triangles.



ANS. FIG. P36.6

P36.7 We assume that she looks only at images in the nearest mirror. The mirrors are 3.00 m apart.

(a) With her palm located 1.00 m in front of the nearest mirror, that she sees its image 1.00 m behind the nearest mirror.

(b) The nearest mirror shows the palm of her hand.

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(c) Her hand is 2.00 m from the farthest mirror, so its image forms 2.00 m behind the farthest mirror, but this image is 2.00 m + 3.00 m = 5.00 m from the nearest mirror, so the image she sees is 5.00 m behind the nearest mirror.

(d) The image is that of the back of her hand reflected in the farthest mirror.

(e) The farthest mirror forms an image of the first image of part (a), which is 1.00 m + 3.00 m = 4.00 m from the farthest mirror; this image is then 4.00 m behind the farthest mirror, so it is 4.00 m + 3.00 m = 7.00 m in front of the nearest mirror, so the image she sees is 7.00 m behind the nearest mirror.

(f) This is the image of the palm reflected back from the nearest to the farthest and back to the nearest mirror.

(g) Since all images are located behind the mirror, and all images result from light reflected in a mirror, all are virtual images.

Section 36.2 Images Formed by Spherical Mirrors

P36.8 (a) A concave mirror is a converging mirror, so the focal length $f = +20.0$ cm. Then, $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$ gives

$$\frac{1}{50.0\text{ cm}}+\frac{1}{q}=\frac{1}{20.0\text{ cm}} \quad \rightarrow \quad q = +33.3\text{ cm}$$

Since $q > 0$, the image is located 33.3 cm in front of the mirror.

(b) $M = -\frac{q}{p} = -\frac{(33.3\text{ cm})}{50.0\text{ cm}} = -0.666$

(c) The image distance is positive, so the image is real.

(d) The magnification is negative, so the image is inverted.

P36.9 We apply the mirror equation using the sign conventions listed in the textbook chapter.

(i) The mirror equation gives

$$\frac{1}{q}=\frac{1}{f}-\frac{1}{p}=\frac{1}{10.0\text{ cm}}-\frac{1}{40.0\text{ cm}} \quad \rightarrow \quad q = 13.3\text{ cm}$$

and $$M=\frac{q}{p}=-\frac{13.3\text{ cm}}{40.0\text{ cm}}=-0.333$$

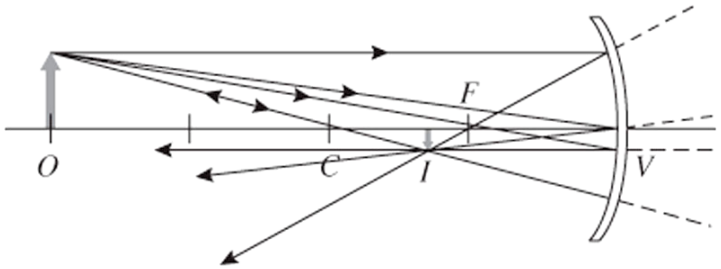
(a) The image is 13.3 cm in front of the mirror.

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(b) The image distance is positive, so the image is real.

(c) The magnification is negative, so the image is inverted.

(d) From above, $M = -0.333$. The value of M indicates that the image is inverted and one-third the height of the object.



ANS. FIG. P36.9(i)

The ray diagram traced in ANS. FIG. P36.9(i) shows this identification more clearly, and that the image is inverted.

(ii) Again, from the mirror equation,

$$\frac{1}{q} = \frac{1}{f} - \frac{1}{p} = \frac{1}{10.0\text{ cm}} - \frac{1}{20.0\text{ cm}} \quad \rightarrow \quad q = 20.0\text{ cm}$$

and

$$M = \frac{q}{p} = -\frac{20.0\text{ cm}}{20.0\text{ cm}} = -1.00$$

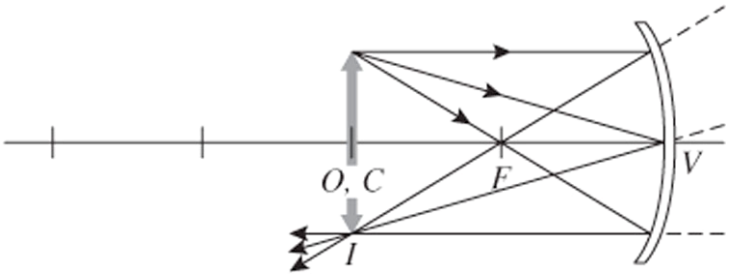
The ray diagram for this case is shown in ANS. FIG. P36.9(ii).

(a) The image is 20.0 cm in front of the mirror.

(b) The image distance is positive, so the image is real.

(c) The magnification is negative, so the image is inverted.

(d) From above, $M = -1.00$. The value of M indicates that the image is inverted and the same height as the object in this special case.



ANS. FIG. P36.9(ii)

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(iii) (a) The object is now at the focal point of the mirror. Following the same steps gives

$$q = \frac{1}{2/R - 1/p} = \frac{1}{2/(20.0\text{ cm}) - 1/(10.0\text{ cm})} = \frac{1}{0} = \infty$$

We can say that no image is formed, or that the image is at an infinite distance. The ray diagram for this case is shown in ANS. FIG. P36.9(iii).

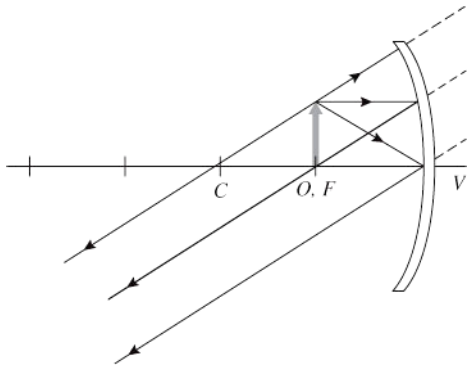
(b) In this special case the reflected rays do not intersect. We cannot classify the image as real or virtual as no image is formed.

(c) We cannot classify the image as upright or inverted as no image is formed.

A screen placed at a large distance in front of the mirror can intercept the reflected light energy, showing the appearance of an upside-down real image, but it is not sharp for any finite distance. You can look into the mirror to view the image as a right side up virtual image, with your eye focused on infinity.

(d) The magnification is $M = -\frac{q}{p} = -\frac{\infty}{20.0\text{ cm}} = \infty$.

In this special case, if we say no image is formed at a finite distance, it has no finite magnification. If we say the image is at infinity, then its height and its magnification are also infinite. There is no physical difference between $+\infty$ and $-\infty$.

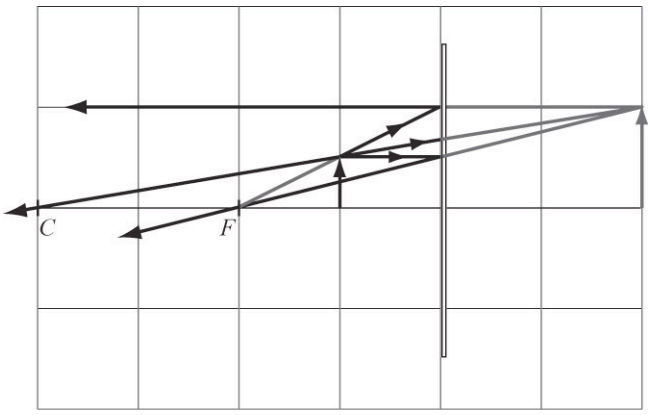


ANS. FIG. P36.9(iii)

P36.10 (a) To approximate paraxial rays, the rays should be drawn so that they reflect at the vertical plane that passes through the vertex of the mirror, rather than at the mirror's surface, as done in the textbook. For this reason, the concave surface of the mirror appears flat in ANS. FIG. P36.10.

(b) q = –40.0 cm, so the image is behind the mirror.

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ANS. FIG. P36.10

(c) $\boxed{M = +2.00, \text{ so the image is enlarged and upright}}$.

(d) The mirror is concave (converging), so $f = +40.0$ cm.

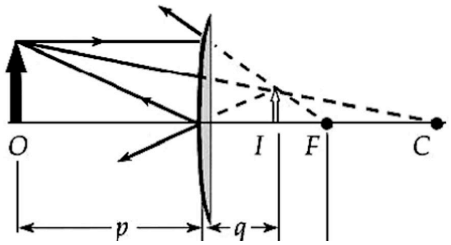
$$\frac{1}{q} = \frac{1}{f} - \frac{1}{p} = \frac{1}{40.0 \text{ cm}} - \frac{1}{20.0 \text{ m}} \quad \rightarrow \quad q = -40.0 \text{ cm}$$

and $$M = \frac{-q}{p} = \frac{-(-40.0 \text{ cm})}{20.0 \text{ cm}} = +2.00$$

P36.11 The convex mirror is described by

$$f = \frac{R}{2} = \frac{-40.0 \text{ cm}}{2} = -20.0 \text{ cm}$$

ANS. FIG. P36.11 shows the ray diagram for this situation.



ANS. FIG. P36.11

(a) Then $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$ gives

$$q = \frac{1}{1/f - 1/p} = \frac{1}{1/(-20.0 \text{ cm}) - 1/(30.0 \text{ cm})} = \boxed{-12.0 \text{ cm}}$$

The magnification factor is

$$M = -\frac{q}{p} = -\left(\frac{-12.0 \text{ cm}}{30.0 \text{ cm}}\right) = \boxed{+0.400}$$

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The image is behind the mirror, upright, virtual, and diminished.

(b) Following the same steps,

$$q = \frac{1}{1/f - 1/p} = \frac{1}{1/(-20.0\text{ cm}) - 1/(60.0\text{ cm})} = \boxed{-15.0\text{ cm}}$$

and $M = \dfrac{-q}{p} = -\left(\dfrac{-15.0\text{ cm}}{60.0\text{ cm}}\right) = \boxed{+0.250}$.

(c) Since $M > 0$, the images are $\boxed{\text{both upright}}$.

P36.12 (a) The mirror is convex (diverging), so

$$f = -\frac{R}{2} = -\frac{0.550\text{ m}}{2} = -0.275\text{ m}$$

$$\frac{1}{q} = \frac{1}{f} - \frac{1}{p} = \frac{1}{-0.275\text{ m}} - \frac{1}{10.0\text{ m}}$$

gives $q = -0.267\text{ m} = \boxed{-26.7\text{ cm}}$.

The image distance is negative; thus, the image is virtual. The image is 26.7 cm behind the mirror.

(b) $M = \dfrac{-q}{p} = -\dfrac{-0.267}{10.0\text{ m}} = +0.026\,7$

The magnification is positive, so the image is $\boxed{\text{upright}}$.

(c) From above, $M = \boxed{0.026\,7}$.

P36.13 (a) The mirror is convex (diverging), so $f = -10.0$ cm.

$$\frac{1}{q} = \frac{1}{f} - \frac{1}{p} = \frac{1}{-10.0\text{ cm}} - \frac{1}{30.0\text{ m}} \quad \rightarrow \quad q = \boxed{-7.50\text{ cm}}$$

The image distance is negative; thus, the image is virtual. The image is 7.50 cm behind the mirror.

(b) From $M = \dfrac{-q}{p} = -\dfrac{-7.50}{30.0\text{ cm}} = +0.250$, we see that the magnification is positive, so the image is $\boxed{\text{upright}}$.

(c) $M = \dfrac{h'}{h} \quad \rightarrow \quad h' = Mh = +0.250(2.00\text{ cm}) = \boxed{0.500\text{ cm}}$

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P36.14 (a) Since the object is in front of the mirror, $p > 0$, and $p = 1.00$ cm. With the image behind the mirror, the image is virtual, so $q < 0$, and $q = -10.0$ cm. The mirror equation gives for the radius of curvature

$$\frac{1}{f} = \frac{2}{R} = \frac{1}{p} + \frac{1}{q} = \frac{1}{1.00 \text{ cm}} + \frac{1}{-10.0 \text{ cm}} \quad \rightarrow \quad f = \frac{R}{2} = 1.11 \text{ cm}$$

$$R = \boxed{+2.22 \text{ cm}}$$

A positive radius means the mirror is converging, so it is a concave mirror.

(b) The magnification is $M = -\frac{q}{p} = -\frac{(-10.0 \text{ cm})}{1.00 \text{ cm}} = \boxed{+10.0}$.

P36.15 The niche acts as a cylindrical mirror that reflects sound. This is a mirror with a vertical axis and a radius $R = 2.50$ m: its focal length $f = \frac{R}{2} = 1.25$ m. To the extent that we can treat sound as being composed of "rays of sound," we can find the point of focus of sound waves by using the same method we use for rays of light.

In a vertical plane the sound disperses as usual, but that radiated in a horizontal plane is concentrated in a sound image at distance q from the back of the niche, where

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \quad \rightarrow \quad \frac{1}{2.00 \text{ m}} + \frac{1}{q} = \frac{1}{1.25 \text{ m}}$$

$$q = \boxed{3.33 \text{ m from the deepest point in the niche}}.$$

P36.16 A convex mirror *diverges* light rays incident upon it, so the mirror in this problem cannot focus the Sun's rays to a point.

P36.17 From the definition of magnification, $M = -\frac{q}{p}$, which gives

$$q = -Mp = -0.013\ 0(30 \text{ cm}) = -0.390 \text{ cm}$$

Then, from the mirror-lens equation,

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{2}{R}$$

$$\frac{1}{30.0 \text{ cm}} + \frac{1}{-0.390 \text{ cm}} = \frac{2}{R}$$

$$R = -0.790 \text{ cm}$$

The cornea is convex, with radius of curvature $\boxed{0.790 \text{ cm}}$.

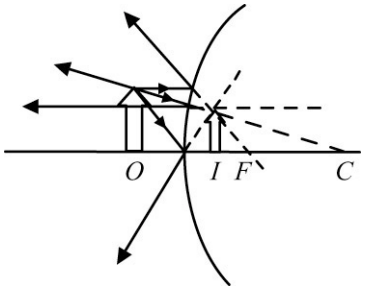
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P36.18 The ball is a convex mirror with a diameter of 8.50 cm:

$$R = -4.25 \text{ cm} \quad \text{and} \quad f = \frac{R}{2} = -2.125 \text{ cm}$$

(a) We have

$$M = \frac{3}{4} = -\frac{q}{p} \quad \rightarrow \quad q = -\frac{3}{4}p$$



ANS. FIG. P36.18

By the mirror equation,

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

$$\frac{1}{p} + \frac{1}{-(3/4)p} = \frac{1}{-2.125 \text{ cm}}$$

or $$\frac{3}{3p} - \frac{4}{3p} = \frac{1}{-2.125 \text{ cm}} = \frac{-1}{3p} \quad \rightarrow \quad p = +0.708 \text{ m}$$

The object is $\boxed{0.708 \text{ m in front of the sphere}}$.

(b) From ANS. FIG. P36.18, the image is $\boxed{\text{upright}}$, virtual, and diminished.

P36.19 (a) The image is inverted and 4.00 times larger, so the magnification is

$$M = -4.00 = -\frac{q}{p} \quad \rightarrow \quad q = 4.00p$$

Thus the image is farther from the mirror than the object.

The object and images distances are related by

$$q - p = 0.600 \text{ m} = 4.00p - p = 3.00p \quad \rightarrow \quad p = 0.200 \text{ m},$$

and $q = 0.800$ m.

By the mirror equation,

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q} = \frac{1}{0.200 \text{ m}} + \frac{1}{0.800 \text{ m}} \quad \rightarrow \quad f = \boxed{0.160 \text{ m}}$$

(b) A convex (diverging) mirror forms an upright, virtual image, so the magnification is

$$M = +0.500 = -\frac{q}{p} \quad \rightarrow \quad q = -0.500p$$

The image is virtual, so it is behind the mirror, and the image distance is negative. The object and images distances are related

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by

$$|q| + p = 0.600 \text{ m} = -q + p = -\left(-0.500p\right) + p = 1.50p$$

$$p = 0.400 \text{ m} \quad \rightarrow \quad q = -0.200 \text{ m}$$

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{1}{0.400 \text{ m}} + \frac{1}{-0.200 \text{ m}} \quad \rightarrow \quad f = \boxed{-0.400 \text{ m}}$$

P36.20 (a) The image is inverted, and $a > 1$ times larger, so the magnification is

$$M = -a = -\frac{q}{p} \quad \rightarrow \quad q = ap$$

Thus the image is farther from the mirror than the object.

The object and image distances are related by

$$q - p = d = ap - p = (a-1)p \quad \rightarrow \quad p = \frac{d}{a-1}, \qquad q = \frac{ad}{a-1}$$

By the mirror equation,

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q} = \frac{a-1}{d} + \frac{a-1}{ad} = \frac{a(a-1)+(a-1)}{ad} = \frac{a^2-1}{ad}$$

$$f = \boxed{\frac{ad}{a^2-1}}$$

(b) The image is upright, and $a < 1$, so the magnification is:

$$M = a = -\frac{q}{p} \quad \rightarrow \quad q = -ap$$

The image is virtual, so it is behind the mirror, and the image distance is negative. The object and image distances are related by

$$|q| + p = d = -q + p = -\left(-ap\right) + p = (a+1)p$$

$$p = \frac{d}{1+a} \qquad \text{and} \qquad q = -ap = \frac{-ad}{1+a}$$

By the mirror equation,

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q} = \frac{1+a}{d} + \frac{1+a}{-ad} = \frac{a(1+a)-(1+a)}{ad} = \frac{a^2-1}{ad}$$

$$f = \boxed{\frac{ad}{a^2-1}}$$

P36.21 From the magnification equation,

$$M = \frac{h'}{h} = \frac{+4.00 \text{ cm}}{10.0 \text{ cm}} = +0.400 = -\frac{q}{p}$$

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which gives $q = -0.400p$, so the image must be virtual.

(a) It is a (diverging) convex mirror that produces a diminished, upright virtual image.

(b) We must have

$$p + |q| = 42.0 \text{ cm} = p - q$$

$$p = 42.0 \text{ cm} + q$$

$$p = 42.0 \text{ cm} - 0.400p$$

$$p = \frac{42.0 \text{ cm}}{1.40} = 30.0 \text{ cm}$$

The mirror is at the 30.0-cm mark.

(c) $$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{1}{30 \text{ cm}} + \frac{1}{-0.400(30 \text{ cm})} = \frac{1}{f} = -0.050\,0/\text{cm}$$

$f = -20.0$ cm

The ray diagram looks like Figure 36.13(c) in the text.

P36.22 (a) Since the mirror is concave, $R > 0$, giving $f = \frac{R}{2} = +12.0$ cm. The magnification is positive because the image is upright:

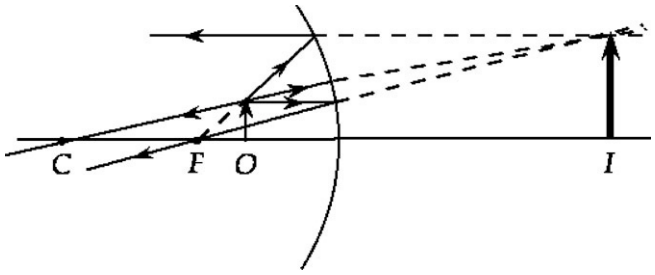
$$M = -\frac{q}{p} = +3 \quad \rightarrow \quad q = -3p$$

The mirror equation is then

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

$$\frac{1}{p} - \frac{1}{3p} = \frac{2}{3p} = \frac{1}{12.0 \text{ cm}} \quad \rightarrow \quad p = \boxed{8.00 \text{ cm}}$$

(b) ANS. FIG. P36.22(b) shows the principal ray diagram for this situation.



ANS. FIG. P36.22(b)

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(c) The image distance is negative, so the image is virtual. The rays of light do not actually come from the position of the image.

P36.23 Assume that the object distance is the same in both cases (i.e., her face is the same distance from the hubcap regardless of which way it is turned). Also realize that the near image (q = –10.0 cm) occurs when using the convex side of the hubcap. Applying the mirror equation to both cases gives:

concave side: $R = |R|$, q = –30.0 cm

$$\frac{1}{p} - \frac{1}{30.0} = \frac{2}{|R|} \quad \text{or} \quad \frac{2}{|R|} = \frac{30.0 \text{ cm} - p}{(30.0 \text{ cm})p} \qquad \textbf{[1]}$$

convex side: $R = -|R|$, q = –10.0 cm

$$\frac{1}{p} - \frac{1}{10.0} = -\frac{2}{|R|} \quad \text{or} \quad \frac{2}{|R|} = \frac{p - 10.0 \text{ cm}}{(10.0 \text{ cm})p} \qquad \textbf{[2]}$$

(a) Equating equations [1] and [2] gives:

$$\frac{30.0 \text{ cm} - p}{3.00} = p - 10.0 \text{ cm} \quad \text{or} \quad p = 15.0 \text{ cm}$$

Thus, her face is 15.0 cm from the hubcap.

(b) Using the above result (p = 15.0 cm) in equation [1] gives:

$$\frac{2}{|R|} = \frac{30.0 \text{ cm} - 15.0 \text{ cm}}{(30.0 \text{ cm})(15.0 \text{ cm})} \quad \text{or} \quad \frac{2}{|R|} = \frac{1}{30.0 \text{ cm}}$$

and $|R| = 60.0$ cm.

The radius of the hubcap is 60.0 cm.

P36.24 (a) We assume the object is real; thus the object distance p is positive. The mirror is convex, so it is a diverging mirror, and we have $f = -|f| = -8.00$ cm. The image is virtual, so $q = -|q|$. Since we also know that $|q| = p/3$, the mirror equation gives

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{p} - \frac{3}{p} = \frac{1}{f} \quad \text{or} \quad -\frac{2}{p} = \frac{1}{-8.00 \text{ cm}}$$

so p = +16.0 cm

This means that the object is 16.0 cm from the mirror.

(b) The magnification is $M = -q/p = +|q|/p = +1/3 =$ +0.333.

(c) Thus, the image is upright and one-third the size of the object.

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P36.25 (a) The image forms on a screen, so it is real and in front of the mirror, so $q = p + 5.00$ m, because p is positive. The magnification is

$$M = -\frac{q}{p} = -5.00 \quad \text{or} \quad q = 5.00p$$

Therefore,

$$p + 5.00 \text{ m} = 5.00p \quad \rightarrow \quad p = 1.25 \text{ m}$$

and $q = p + 5.00 \text{ m} = 6.25$ m. From

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{1}{1.25 \text{ m}} + \frac{1}{6.25 \text{ m}} \quad \rightarrow \quad f = +1.04 \text{ m}$$

The focal length is positive, so the mirror is a converging mirror: concave.

(b) $f = +1.04 \text{ m} = \dfrac{R}{2} \rightarrow R =$ 2.08 m

(c) From part (a), $p = 1.25$ m; the mirror should be 1.25 m from the object.

P36.26 (a) The image starts from a point whose height above the mirror vertex is given by

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{2}{R}$$

$$\frac{1}{3.00 \text{ m}} + \frac{1}{q} = \frac{1}{0.500 \text{ m}} \quad \rightarrow \quad q = 0.600 \text{ m}$$

As the ball falls, p decreases and q increases. Ball and image pass when $q_1 = p_1$. When this is true,

$$\frac{1}{p_1} + \frac{1}{p_1} = \frac{1}{0.500 \text{ m}} = \frac{2}{p_1} \quad \rightarrow \quad p_1 = 1.00 \text{ m},$$

which is at the focal point.

As the ball passes the focal point, the image switches from infinitely far above the mirror to infinitely far below the mirror. As the ball approaches the mirror from above, the virtual image approaches the mirror from below, reaching it together when $p_2 = q_2 = 0$.

(b) The falling ball passes its real image when it has fallen

$$\Delta y = 3.00 \text{ m} - 1.00 \text{ m} = 2.00 \text{ m} = \frac{1}{2} g t^2$$

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which gives $t = \sqrt{\frac{2(2.00\text{ m})}{9.80\text{ m/s}^2}} = \boxed{0.639\text{ s}}$.

The ball reaches its virtual image when it reaches the surface of the mirror, which is when it has traversed

$$\Delta y = 3.00\text{ m} - 0 = 3.00\text{ m} = \frac{1}{2}gt^2$$

which gives $t = \sqrt{\frac{2(3.00\text{ m})}{9.80\text{ m/s}^2}} = \boxed{0.782\text{ s}}$.

P36.27 (a) The flat mirror produces an image according to $q = -p = -24.0$ cm. The image is behind the mirror, with the distance from your eyes given by

$$1.55\text{ m} + 24.0\text{ m} = \boxed{25.6\text{ m}}$$

(b) The image is the same size as the object, so

$$\theta = \frac{h}{d} = \frac{1.50\text{ m}}{25.6\text{ m}} = \boxed{0.058\ 7\text{ rad}}$$

(c) $$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{2}{R}$$

$$\frac{1}{24\text{ m}} + \frac{1}{q} = \frac{2}{(-2\text{ m})} \quad \rightarrow \quad q = -0.960\text{ m}$$

This image is behind the mirror, distant from your eyes by

$$1.55\text{ m} + 0.960\text{ m} = \boxed{2.51\text{ m}}$$

(d) The image size is given by $M = \frac{h'}{h} = -\frac{q}{p}$:

$$h' = -h\frac{q}{p} = -1.50\text{ m}\left(\frac{-0.960\text{ m}}{24\text{ m}}\right) = 0.060\ 0\text{ m}$$

So its angular size at your eye is $\theta' = \frac{h'}{d} = \frac{0.06\text{ m}}{2.51\text{ m}} = \boxed{0.023\ 9\text{ rad}}$.

(e) Your brain assumes that the car is 1.50 m high and calculates its distance as

$$d' = \frac{h}{\theta'} = \frac{1.50\text{ m}}{0.023\ 9} = \boxed{62.8\text{ m}}$$

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***P36.28** The focal length of the mirror may be found from the given object and image distances as

$$\frac{1}{f}=\frac{1}{p}+\frac{1}{q}$$

Solving for the focal length f gives

$$f=\frac{pq}{p+q}=\frac{(152\text{ cm})(18.0\text{ cm})}{152\text{ cm}+18.0\text{ cm}}=+16.1\text{ cm}$$

For an upright image twice the size of the object, the magnification is

$$M=-\frac{q}{p}=+2.00$$

which gives $q = -2.00p$.

Then, using the mirror equation again, $\frac{1}{f}=\frac{1}{p}+\frac{1}{q}$ becomes

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{p}-\frac{1}{2.00p}=\frac{2-1}{2.00p}=\frac{1}{f}$$

or $$p=\frac{f}{2.00}=\frac{16.1\text{ cm}}{2.00}=\boxed{8.05\text{ cm}}$$

Section 36.3 Images Formed by Refraction

P36.29 The image forms within the rod.

$$\frac{n_1}{p}+\frac{n_2}{q}=\frac{n_2-n_1}{R}\quad\rightarrow\quad\frac{1.00}{p}+\frac{1.50}{q}=\frac{1.50-1.00}{6.00\text{ cm}}=\frac{1}{12.0\text{ cm}}$$

(a) $$\frac{1.00}{20.0\text{ cm}}+\frac{1.50}{q}=\frac{1}{12.0\text{ cm}}\quad\rightarrow\quad q=\boxed{45.0\text{ cm}}$$

(b) $$\frac{1.00}{10.0\text{ cm}}+\frac{1.50}{q}=\frac{1}{12.0\text{ cm}}\quad\rightarrow\quad q=\boxed{-90.0\text{ cm}}$$

(c) $$\frac{1.00}{3.0\text{ cm}}+\frac{1.50}{q}=\frac{1}{12.0\text{ cm}}\quad\rightarrow\quad q=\boxed{-6.00\text{ cm}}$$

P36.30 $$\frac{n_1}{p}+\frac{n_2}{q}=\frac{n_2-n_1}{R}=0\quad\text{and}\quad R\rightarrow\infty$$

$$q=-\frac{n_2}{n_1}p=-\frac{1}{1.309}(50.0\text{ cm})=-38.2\text{ cm}$$

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Thus, the virtual image of the dust speck is
$\boxed{38.2 \text{ cm below the top surface}}$ of the ice.

P36.31 For a plane refracting (water) surface $(R \to \infty)$

$$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R} \quad \text{becomes} \quad q = -\frac{n_2}{n_1} p$$

(a) When the pool is full, $p = 2.00$ m and

$$q = -\left(\frac{1.00}{1.333}\right)(2.00 \text{ m}) = -1.50 \text{ m}$$

or the pool appears to be $\boxed{1.50 \text{ m}}$ deep.

(b) If the pool is half filled, then $p = 1.00$ m and $q = -0.750$ m. Thus, the bottom of the pool appears to be 0.750 m below the water surface or $\boxed{1.75 \text{ m}}$ below ground level.

P36.32 Since the center of curvature of the surface is on the side the light comes from, $R < 0$ giving $R = -4.00$ cm. For the line, $p = 4.00$ cm; then,

$$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$$

becomes

$$\frac{1.00}{q} = \frac{1.00 - 1.50}{-4.00 \text{ cm}} - \frac{1.50}{4.00 \text{ cm}}$$

or $q = -4.00$ cm

Thus, the magnification $M = \frac{h'}{h} = -\left(\frac{n_1}{n_2}\right)\frac{q}{p}$ gives

$$h' = -\left(\frac{n_1 q}{n_2 p}\right)h = -\frac{1.50(-4.00 \text{ cm})}{1.00(4.00 \text{ cm})}(2.50 \text{ mm}) = \boxed{3.75 \text{ mm}}$$

P36.33 The water's surface has no curvature. When $R \to \infty$, the equation $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$, which describes image formation at a single refracting surface, becomes $q = -p\left(\frac{n_2}{n_1}\right)$. We use this to locate the final images of the two surfaces of the glass plate. First, find the image the glass forms of the *bottom* of the plate. [From Table 35.1, for flint glass, $n_1 = 1.66$.]

$$q_{B1} = -\left(\frac{1.33}{1.66}\right)(8.00 \text{ cm}) = -6.41 \text{ cm}$$

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This virtual image is 6.41 cm below the top surface of the glass, or 18.41 cm below the water surface. Next, use this image as an object and locate the image the water forms of the bottom of the plate.

$$q_{B2} = -\left(\frac{1.00}{1.33}\right)(18.41 \text{ cm}) = -13.84 \text{ cm}$$

$$= 13.84 \text{ cm below the water surface}$$

Now find image the water forms of the *top* surface of the glass.

$$q_3 = -\left(\frac{1}{1.33}\right)(12.0 \text{ cm}) = -9.02 \text{ cm}$$

$$= 9.02 \text{ cm below the water surface}$$

Therefore, the apparent thickness of the glass is

$$\Delta t = 13.84 \text{ cm} - 9.02 \text{ cm} = \boxed{4.82 \text{ cm}}$$

P36.34 Refer to Figure P36.34 in the textbook. In the right triangle lying between O and the center of the curved surface, $\tan\theta_1 = h/p$. In the right triangle lying between I and the center of the surface, $\tan\theta_2 = -h'/q$. We need the negative sign because the image height is counted as negative while the angle is not. We substitute into the given

$$n_1 \tan\theta_1 = n_2 \tan\theta_2$$

to obtain

$$n_1 h/p = -n_2 h'/q$$

Then the magnification, defined by $M = h'/h$, is given by

$$M = h'/h = -n_1 q/n_2 p$$

P36.35 From Equation 36.8 for image formation by a single refracting surface,

$$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$$

We solve for q to find

$$q = \frac{n_2 R p}{p(n_2 - n_1) - n_1 R}.$$

In this case, $n_1 = 1.50$, $n_2 = 1.00$, $p = 10.0$ cm, and $R = -15.0$ cm.

So the image location is

$$q = \frac{(1.00)(-15.0 \text{ cm})(10.0 \text{ cm})}{(10.0 \text{ cm})(1.00 - 1.50) - (1.50)(-15.0 \text{ cm})} = -8.57 \text{ cm}$$

$\boxed{\text{apparent depth is 8.57 cm}}$

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P36.36 The center of curvature is on the object side, so the radius of curvature is negative: $R = -|R| = -225$ cm.

(a) (i) $p = 5.00$ cm:

$$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$$

$$\frac{1.333}{5.00\text{ cm}} + \frac{1.000}{q} = \frac{1.000 - 1.333}{-225\text{ cm}} \quad \rightarrow \quad q = -3.77\text{ cm}$$

The image is virtual and 3.77 cm from the front wall, in the water.

(ii) $p = 25.0$ cm:

$$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$$

$$\frac{1.333}{25.0\text{ cm}} + \frac{1.000}{q} = \frac{1.000 - 1.333}{-225\text{ cm}} \quad \rightarrow \quad q = -19.3\text{ cm}$$

The image is virtual and 19.3 cm from the front wall, in the water.

(b) From Problem 34, the magnification is $M = -\dfrac{n_1 q}{n_2 p}$.

(i) $$M = -\frac{n_1 q}{n_2 p} = -\frac{1.333(-3.77\text{ cm})}{1.00(5.00\text{ cm})} = \boxed{+1.01}$$

(ii) $$M = -\frac{n_1 q}{n_2 p} = -\frac{1.333(-19.2\text{ cm})}{1.000(25.0\text{ cm})} = \boxed{+1.03}$$

(c) The plastic has uniform thickness, so the surfaces of entry and exit for any particular ray are very nearly parallel. The ray is slightly displaced, but it would not be changed in direction by going through the plastic wall with air on both sides. Only the difference between the air and water is responsible for the refraction of the light.

(d) Yes

(e) If $p = |R|$, from $\dfrac{n_1}{p} + \dfrac{n_2}{q} = \dfrac{n_2 - n_1}{R} = \dfrac{n_1 - n_2}{|R|}$ we have

$$\frac{n_1}{|R|} + \frac{n_2}{q} = \frac{n_1 - n_2}{|R|} \quad \rightarrow \quad \frac{n_2}{q} = \frac{-n_2}{|R|}$$

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then $q = -|R|$.

If $p > |R|$ (but also $p < 4.00|R|$, if the image is to be virtual—see **NOTE** below), then

$$p > |R| \quad \rightarrow \quad \frac{1}{|R|} > \frac{1}{p} \quad \rightarrow \quad \frac{1}{|R|} - \frac{1}{p} > 0$$

and

$$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_1 - n_2}{|R|}$$

$$\frac{n_2}{q} = \frac{n_1}{|R|} - \frac{n_2}{|R|} - \frac{n_1}{p}$$

$$\frac{1}{q} = -\frac{1}{|R|} + \frac{n_1}{n_2}\left(\frac{1}{|R|} - \frac{1}{p}\right)$$

$$\frac{1}{q} = -\frac{1}{|R|} + (1.333)\left(\frac{1}{|R|} - \frac{1}{p}\right)$$

$$\frac{1}{|q|} = \frac{1}{|R|} - (1.333)\left(\frac{1}{|R|} - \frac{1}{p}\right) < \frac{1}{|R|} \quad \rightarrow \quad |q| > |R|$$

[Assuming that $p < 4.00|R|$.] For example, if $p = 2|R|$,

$$\frac{1}{q} = -\frac{1}{|R|} + (1.333)\left(\frac{1}{|R|} - \frac{1}{2|R|}\right) = \frac{1}{|R|}\left(-1 + \frac{1.333}{2}\right) = \frac{-0.3335}{|R|}$$

$$q = -3.00|R|$$

$$M = -\frac{n_1 q}{n_2 p} = -\frac{1.333\left(-3.00|R|\right)}{1.000\left(2|R|\right)} = +2.00$$

Summarizing our results:

If $p = \|R\|$, then $q = -p = -\|R\|$; if $p > \|R\|$, then $\|q\| > \|R\|$. For example, if $p = 2\|R\|$, then $q = -3.00\|R\|$ and $M = +2.00$.

NOTE: In the equation $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R} = \frac{n_1 - n_2}{|R|}$, the term $\frac{n_1 - n_2}{|R|}$ is positive because $n_1 > n_2$. If the image is to be virtual, then q must be negative, and so the term $(n_1 - n_2)/|R|$ must be less than n_1/p:

$$\frac{n_1 - n_2}{|R|} < \frac{n_1}{p} \quad \rightarrow \quad p < \frac{n_1}{n_1 - n_2}|R| = \frac{1.333}{1.333 - 1.000}|R| = 4.00|R|$$

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P36.37 For a plane surface ($R = \infty$), $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$ becomes $q = -\frac{n_2 p}{n_1}$.

Thus, the magnitudes of the rate of change in the image and object positions are related by

$$\left|\frac{dq}{dt}\right| = \frac{n_2}{n_1}\left|\frac{dp}{dt}\right|$$

If the fish swims toward the wall with a speed of 2.00 cm/s, the speed of the image is given by

$$v_{\text{image}} = \left|\frac{dq}{dt}\right| = \frac{1.00}{1.33}(2.00 \text{ cm/s}) = \boxed{1.50 \text{ cm/s}}$$

Section 36.4 Images Formed by Thin Lenses

***P36.38** (a) From $\frac{1}{q} = \frac{1}{f} - \frac{1}{p} = \frac{1}{25.0 \text{ cm}} - \frac{1}{26.0 \text{ cm}}$, we obtain

$\boxed{q = 650 \text{ cm}}$.

The image is $\boxed{\text{real, inverted, and enlarged}}$.

(b) From $\frac{1}{q} = \frac{1}{f} - \frac{1}{p} = \frac{1}{25.0 \text{ cm}} - \frac{1}{24.0 \text{ cm}}$, we obtain

$\boxed{q = -600 \text{ cm}}$.

The image is $\boxed{\text{virtual, upright, and enlarged}}$.

***P36.39** (a) From the mirror-and-lens equation,

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}: \qquad \frac{1}{32.0 \text{ cm}} + \frac{1}{8.00 \text{ cm}} = \frac{1}{f}$$

so $\boxed{f = 6.40 \text{ cm}}$.

(b) $M = -\frac{q}{p} = -\frac{8.00 \text{ cm}}{32.0 \text{ cm}} = \boxed{-0.250}$

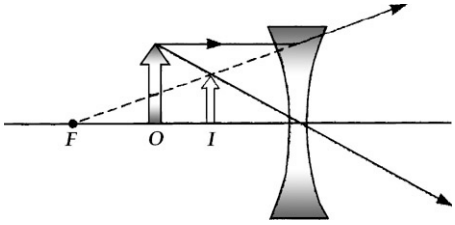
(c) Since $f > 0$, the lens is $\boxed{\text{converging}}$.

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P36.40 (a) From the mirror-and-lens equation, $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$:

$$\frac{1}{20.0\text{ cm}}+\frac{1}{q}=\frac{1}{(-32.0\text{ cm})}$$

$$\rightarrow \quad q=-\left(\frac{1}{20.0}+\frac{1}{32.0}\right)^{-1}=-12.3\text{ cm}$$



ANS. FIG. P36.40

Refer to ANS. FIG. P36.40. The image distance is negative, hence the image is virtual; thus, it forms $\boxed{12.3\text{ cm to the left of the lens}}$.

(b) $M=-\frac{q}{p}=-\frac{(-12.3\text{ cm})}{20.0\text{ cm}}=\boxed{0.615}$

(c) See the ray diagram shown in ANS. FIG. P36.40.

P36.41 The image is inverted:

$$M=\frac{h'}{h}=\frac{-1.80\text{ m}}{0.024\ 0\text{ m}}=-75.0 \ \rightarrow \ q=75.0p$$

The distance from slide to screen $d = p + q = 3.00$ m:

$$d=p+q=p+75.0p=76.0p$$

$$p=\frac{d}{76.0}=\frac{3.00\text{ m}}{76.0}=0.039\ 5\text{ m}$$

$$p = 39.5\text{ mm}$$

and $q = 75.0p = 2.96$ m.

(a) $\frac{1}{f}=\frac{1}{p}+\frac{1}{q}=\frac{1}{0.039\ 5\text{ m}}+\frac{1}{2.96\text{ m}} \ \rightarrow \ f=0.039\ 0\text{ m}=\boxed{39.0\text{ mm}}$

(b) From above, $p = \boxed{39.5\text{ mm}}$.

P36.42 (a) We are told that $p = 5f$. From the thin lens equation, $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$, we have

$$\frac{1}{5.00f}+\frac{1}{q}=\frac{1}{f} \quad \rightarrow \quad \frac{1}{q}=\frac{1}{f}-\frac{1}{5.00f}=\frac{4.00}{5.00f}$$

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or $$q = \frac{5.00}{4.00} f = +1.25 f$$

The image distance is positive, hence the image is real.

The image is in back of the lens at a distance of $1.25f$ from the lens.

(b) $$M = -\frac{q}{p} = -\frac{1.25 f}{5.00 f} = \boxed{-0.250}$$

(c) From part (a), the image distance is positive, hence the image is real.

P36.43 Let R_1 = outer radius and R_2 = inner radius:

$$\frac{1}{f} = (n-1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = (1.50-1)\left[\frac{1}{2.00\text{ m}} - \frac{1}{2.50\text{ cm}}\right]$$

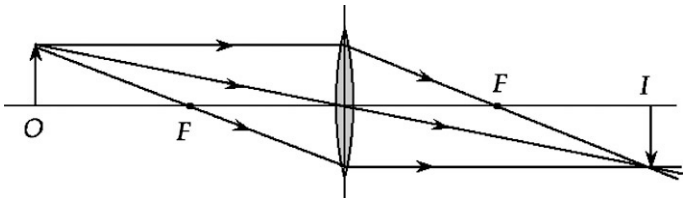
$$= 0.050\ 0\ \text{cm}^{-1}$$

so $f =$ 20.0 cm.

P36.44 Your scale drawings should look similar to those given below:

(i) See diagram in ANS. FIG. P36.44(i).

(a) A carefully drawn-to-scale version of ANS FIG. P36.44(i) should yield an inverted image 20.0 cm in back of the lens and the same size as the object.



ANS. FIG. P36.44(i)

(b) The image forms behind the lens, so the image is real.

(c) The figure shows that the image is inverted.

(d) The height of the image is the same as the height of the object, so $M = -1.00$.

(e) $$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}: \quad \frac{1}{20.0\,\text{cm}} + \frac{1}{q} = \frac{1}{10.0\,\text{cm}} \rightarrow q = +20.0\,\text{cm}$$

A positive image distance means that the image is real.

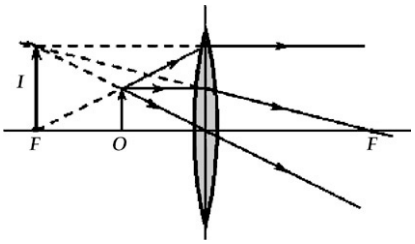
The magnification is $$M = -\frac{q}{p} = -\frac{+20.0\,\text{cm}}{20.0\,\text{cm}} = -1.00$$

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A negative magnification means that the image is inverted.

Algebraic answers agree, and we can express values to three significant figures: $q = 20.0$ cm, $M = -1.00$.

(ii) See diagram in ANS. FIG. P36.44(ii).



ANS. FIG. P36.44(ii)

(a) A carefully drawn-to-scale version of ANS FIG. P36.44(ii) should yield an upright, virtual image located 10 cm in front of the lens and twice the size of the object.

(b) The image forms in front of the lens, so the image is virtual.

(c) The figure shows that the image is upright.

(d) The height of the image is twice that of the object, so $M = +2.00$.

(e) $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$: $\quad \frac{1}{5.00\text{ cm}}+\frac{1}{q}=\frac{1}{10.0\text{ cm}} \rightarrow q = -10.0$ cm

A negative image distance means that the image is virtual.

The magnification is $M = -\frac{q}{p} = -\frac{(-10.0\text{ cm})}{5.00\text{ cm}} = +2.00$

A positive magnification means that the image is upright.

Algebraic answers agree, and we can express values to three significant figures: $q = -10.0$ cm, $M = +2.00$.

(f) Small variations from the correct directions of rays can lead to significant errors in the intersection point of the rays. These variations may lead to the three principal rays not intersecting at a single point.

P36.45 In parts (a) and (b), the images are real, so the image distances are positive.

(a) $q = +20.0$ cm:

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}: \quad \frac{1}{p}+\frac{1}{20.0\text{ cm}}=\frac{1}{10.0\text{ cm}} \quad \rightarrow \quad p = +20.0\text{ cm}$$

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The object distance is positive, so the object is real.

The object is $\boxed{20.0 \text{ cm from the lens on the front side}}$.

(b) $q = +50.0$ cm:

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}: \quad \frac{1}{p}+\frac{1}{50.0 \text{ cm}}=\frac{1}{10.0 \text{ cm}}$$

$$p = +12.5 \text{ cm}$$

The object distance is positive, so the object is real.

The object is $\boxed{12.5 \text{ cm from the lens on the front side}}$.

(c and d) Now, the images in parts (a) and (b) are virtual, so the image distances are negative.

(c) $q = -20.0$ cm:

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}: \quad \frac{1}{p}+\frac{1}{-20.0 \text{ cm}}=\frac{1}{10.0 \text{ cm}}$$

$$p = +6.67 \text{ cm}$$

The object distance is positive, so the object is real.

The object is $\boxed{6.67 \text{ cm from the lens on the front side}}$.

(d) $q = -50.0$ cm:

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}: \quad \frac{1}{p}+\frac{1}{-50.0 \text{ cm}}=\frac{1}{10.0 \text{ cm}}$$

$$p = +8.33 \text{ cm}$$

The object distance is positive, so the object is real.

The object is $\boxed{8.33 \text{ cm from the lens on the front side}}$.

P36.46 Use the thin lens equation, $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$. The magnification is $M=-\frac{q}{p}$.

(i) $p = 40.0$ cm: $\quad \frac{1}{40.0 \text{ cm}}+\frac{1}{q}=\frac{1}{-20.0 \text{ cm}} \quad \rightarrow \quad q = -13.3$ cm

$$M=-\frac{q}{p}=-\frac{(-13.3 \text{ cm})}{40.0 \text{ cm}}=+0.333$$

(a) The image forms $\boxed{13.3 \text{ cm in front of the lens}}$.

(b) The object distance is negative, so the image is $\boxed{\text{virtual}}$.

(c) The magnification is positive, so the image is $\boxed{\text{upright}}$.

(d) From above, $M = \boxed{+0.333}$

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(ii) $p = 20.0$ cm: $\dfrac{1}{20.0\text{ cm}} + \dfrac{1}{q} = \dfrac{1}{-20.0\text{ cm}} \rightarrow q = -10.0\text{ cm}$

$$M = -\frac{q}{p} = -\frac{(-10.0\text{ cm})}{20.0\text{ cm}} = +0.500$$

(a) The image forms $\boxed{10.0\text{ cm in front of the lens}}$.

(b) The object distance is negative, so the image is $\boxed{\text{virtual}}$.

(c) The magnification is positive, so the image is $\boxed{\text{upright}}$.

(d) From above, $M = \boxed{+0.500}$

(iii) $p = 10.0$ cm: $\dfrac{1}{10.0\text{ cm}} + \dfrac{1}{q} = \dfrac{1}{-20.0\text{ cm}} \rightarrow q = -6.67\text{ cm}$

$$M = -\frac{q}{p} = -\frac{(-6.67\text{ cm})}{10.0\text{ cm}} = +0.667$$

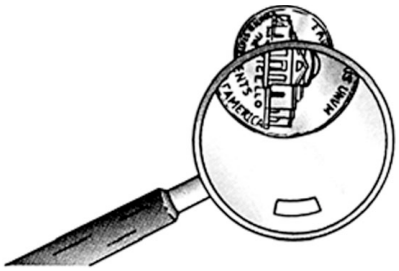
(a) The image forms $\boxed{6.67\text{ cm in front of the lens}}$.

(b) The object distance is negative, so the image is $\boxed{\text{virtual}}$.

(c) The magnification is positive, so the image is $\boxed{\text{upright}}$.

(d) From above, $M = \boxed{+0.667}$

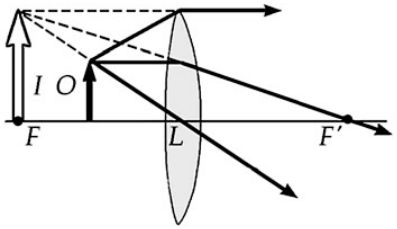
P36.47 We are looking at an enlarged, upright, virtual image. Therefore, $M = +2$ and not -2. Looking through the lens, you see the image beyond the lens. Therefore, the image is virtual, with $q = -2.84$ cm.



ANS. FIG. P36.47(a)

Now, $M = \dfrac{h'}{h} = 2 = -\dfrac{q}{p}$

so $p = -\dfrac{q}{2} = 1.42\text{ cm}$



ANS. FIG. P36.47(b)

A check is that p is positive, as it must be for a real object.

Thus,

$$f = \left(\frac{1}{p} + \frac{1}{q}\right)^{-1}$$

$$= \left[\frac{1}{1.42\text{ cm}} + \frac{1}{(-2.84\text{ cm})}\right]^{-1} = \boxed{2.84\text{ cm}}$$

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P36.48 From the thin lens equation, since the focal length of the lens is constant,

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}:\ p^{-1}+q^{-1}=\text{constant}$$

Differentiating both sides with respect to ρ then gives

$$-1p^{-2}-1q^{-2}\frac{dq}{dp}=0$$

$$\frac{dq}{dp}=-\frac{q^2}{p^2}\quad\rightarrow\quad \boxed{dq=-\frac{q^2}{p^2}dp}$$

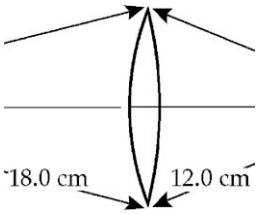
P36.49 We apply the lens maker's equation. The centers of curvature of the lens surfaces are on opposite sides, so the second surface has a negative radius

(a) $$\frac{1}{f}=(n-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]=(0.440)\left[\frac{1}{12.0\text{ cm}}-\frac{1}{(-18.0\text{ cm})}\right]$$

$$f=\boxed{16.4\text{ cm}}$$

(b) $$\frac{1}{f}=(0.440)\left[\frac{1}{18.0\text{ cm}}-\frac{1}{(-12.0\text{ cm})}\right]$$

$$f=\boxed{16.4\text{ cm}}$$



ANS. FIG. P36.49

P36.50 (a) $$\frac{1}{p_a}+\frac{1}{q_a}=\frac{1}{f}\ \text{ becomes }\ \frac{1}{30.0\text{ cm}}+\frac{1}{q_a}=\frac{1}{14.0\text{ cm}}\ \rightarrow\ \boxed{q_a=26.3\text{ cm}}$$

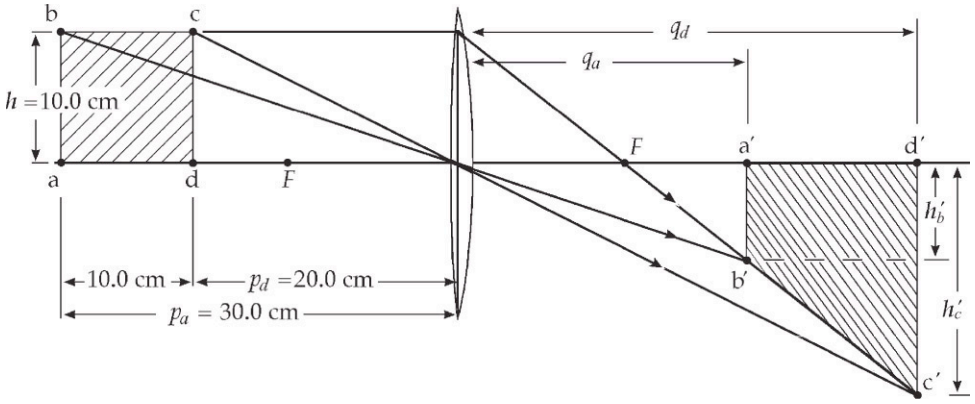
$$\frac{1}{p_d}+\frac{1}{q_d}=\frac{1}{f}\ \text{ becomes }\ \frac{1}{20.0\text{ cm}}+\frac{1}{q_d}=\frac{1}{14.0\text{ cm}}\ \rightarrow\ \boxed{q_d=46.7\text{ cm}}$$

$$h_b'=hM_a=h\left(\frac{-q_a}{p_a}\right)=(10.0\text{ cm})\left(\frac{-26.3\text{ cm}}{30.0\text{ cm}}\right)=\boxed{-8.75\text{ cm}}$$

$$h_c'=hM_d=h\left(\frac{-q_d}{p_d}\right)=(10.0\text{ cm})\left(\frac{-46.7\text{ cm}}{20.0\text{ cm}}\right)=\boxed{-23.3\text{ cm}}$$

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(b) See ANS. FIG. P36.50(b).



ANS. FIG. P36.50(b)

The square is imaged as a trapezoid.

(c) The equation follows from $h'/h = -q/p$ and $1/p + 1/q = 1/f$.

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f} \quad \text{becomes} \quad \frac{1}{p}+\frac{1}{q}=\frac{1}{14\text{ cm}} \quad \text{or} \quad \frac{1}{p}=\frac{1}{14\text{ cm}}-\frac{1}{q}.$$

$$|h'| = |hM| = \left|h\left(\frac{-q}{p}\right)\right| = (10.0\text{ cm})q\left(\frac{1}{14\text{ cm}}-\frac{1}{q}\right)$$

(d) The integral stated adds up the areas of ribbons covering the whole image, each with vertical dimension $|h'|$ and horizontal width dq.

(e) We have

$$\int_{q_a}^{q_d}|h'|\,dq = (10.0\text{ cm})\left(\frac{q^2}{28.0\text{ cm}}-q\right)\Bigg|_{26.3\text{ cm}}^{46.7\text{ cm}}$$

$$= (10.0\text{ cm})\left[\frac{(46.7\text{ cm})^2-(26.3\text{ cm})^2}{28.0\text{ cm}}-46.7\text{ cm}+26.3\text{ cm}\right]$$

$$= \boxed{328\text{ cm}^2}$$

P36.51 In $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$ or $p^{-1}+q^{-1}=$ constant, we differentiate with respect to time:

$$-1\left(p^{-2}\right)\frac{dp}{dt}-1\left(q^{-2}\right)\frac{dq}{dt}=0$$

$$\frac{dq}{dt}=\frac{-q^2}{p^2}\frac{dp}{dt}$$

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We must find the momentary image location q:

$$\frac{1}{20.0\text{ m}}+\frac{1}{q}=\frac{1}{0.300\text{ m}}$$

$$q = 0.305\text{ m}$$

Now $$\frac{dq}{dt}=-\frac{(0.305\text{ m})^2}{(20.0\text{ m})^2}(5.00\text{ m/s})=-0.001\,16\text{ m/s}=1.16\text{ mm/s}.$$

(a) The speed is $\boxed{1.16\text{ mm/s}}$.

(b) Increasing q is away from the lens, negative q is $\boxed{\text{toward the lens}}$. The motion of the image is towards the lens because dq/dt is negative.

P36.52 Let the object distance be p. Then the image distance is $d - p$. Set up the lens equation:

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}\ \rightarrow\ \frac{1}{p}+\frac{1}{d-p}=\frac{1}{f}$$

Rearrange the equation to generate the following quadratic equation:

$$p^2 - dp + df = 0$$

Solve with the quadratic formula:

$$p=\frac{d\pm\sqrt{d^2-4df}}{2} \qquad \textbf{[1]}$$

Substitute numerical values:

$$p=\frac{2.00\text{ m}\pm\sqrt{(2.00\text{ m})^2-4(2.00\text{ m})(0.600\text{ m})}}{2}$$

$$=\frac{2.00\text{ m}\pm\sqrt{-0.800\text{ m}^2}}{2}$$

This expression has no real solutions. Therefore, we cannot find even one position between the object and the screen at which an image is formed on the screen. From equation [1], we see that a real value of p will result only if $d^2 > 4df$, or $d > 4f$, in which case the plus/minus sign in equation [1] will give us two real values for p.

***P36.53** From the thin lens equation, $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$, we obtain

$$q_1=\frac{p_1f_1}{p_1-f_1}=\frac{(4.00\text{ cm})(8.00\text{ cm})}{4.00\text{ cm}-8.00\text{ cm}}=-8.00\text{ cm}$$

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The magnification by the first lens is

$$M_1 = -\frac{q_1}{p_1} = -\frac{(-8.00\text{ cm})}{4.00\text{ cm}} = +2.00$$

The virtual image formed by the first lens is the object for the second lens, so

$$p_2 = 6.00\text{ cm} + |q_1| = 6.00\text{ cm} + 8.00\text{ cm} = +14.00\text{ cm}$$

and the thin lens equation gives

$$q_2 = \frac{p_2 f_2}{p_2 - f_2} = \frac{(14.0\text{ cm})(-16.0\text{ cm})}{14.0\text{ cm} - (-16.0\text{ cm})} = -7.47\text{ cm}$$

The magnification by the second lens is

$$M_2 = -\frac{q_2}{p_2} = -\frac{(-7.47\text{ cm})}{14.0\text{ cm}} = +0.533$$

so the overall magnification is

$$M = M_1 M_2 = (+2.00)(+0.533) = +1.07$$

The position of the final image is 7.47 cm in front of the second lens, and its height is

$$h' = Mh = M_1 M_2 = (+1.07)(1.00\text{ cm}) = \boxed{1.07\text{ cm}}$$

Since $M > 0$, the final image is upright, and since $q_2 < 0$, this image is virtual.

Section 36.5 Lens Abberations

P36.54 Rays from a very distant object are effectively parallel, and the lens is diverging; therefore, the image is virtual and forms at the focal point.

(a) The focal length of the lens is given by

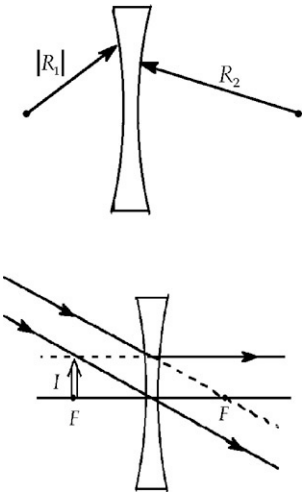
$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.53 - 1.00)\left(\frac{1}{-32.5\text{ cm}} - \frac{1}{42.5\text{ cm}}\right)$$

$$f = -34.7\text{ cm}$$

Note that R_1 is negative because the center of curvature of the first surface is on the virtual image side.

The violet image forms at $\boxed{-34.7\text{ cm}}$

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ANS FIG. P36.54

(b) For red light,

$$\frac{1}{f} = (1.51 - 1.00)\left(\frac{1}{-32.5\text{ cm}} - \frac{1}{42.5\text{ cm}}\right)$$

$$f = -36.1\text{ cm}$$

The red image forms at $\boxed{-36.1\text{ cm}}$.

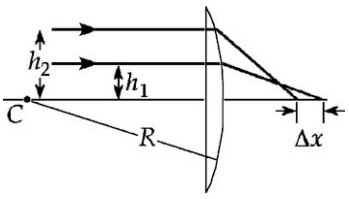
P36.55 Ray h_1 is undeviated at the plane surface and strikes the second surface at angle of incidence given by

$$\theta_1 = \sin^{-1}\left(\frac{h_1}{R}\right) = \sin^{-1}\left(\frac{0.500\text{ cm}}{20.0\text{ cm}}\right) = 1.43°$$

Then,

$$1.00\sin\theta_2 = 1.60\sin\theta_1 = (1.60)\left(\frac{0.500}{20.0\text{ cm}}\right)$$

$$\theta_2 = 2.29°$$



ANS. FIG. P36.55

The angle this emerging ray makes with the horizontal is $\theta_2 - \theta_1 = 0.860°$.

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The ray crosses the axis at a point farther out by f_1 (the focal length):

$$f_1 = \frac{h_1}{\tan(\theta_2 - \theta_1)} = \frac{0.500 \text{ cm}}{\tan(0.860°)} = 33.3 \text{ cm}$$

Because of the curved surface of the lens, the point of exit for this ray is horizontally slightly to the left of the lens vertex (where the principal axis intersects the curved surface of the lens), by the distance

$$R(1 - \cos\theta_1) = 20.0 \text{ cm}[1 - \cos(1.43°)] = 0.006\,25 \text{ cm}$$

Therefore, ray h_1 crosses the axis at this distance from the vertex:

$$x_1 = f_1 - R(1 - \cos\theta_1) = 33.3 \text{ cm} - 0.006\,25 \text{ cm} = 33.3 \text{ cm}$$

Now we repeat the above calculation for ray h_2:

$$\theta = \sin^{-1}\left(\frac{12.0 \text{ cm}}{20.0 \text{ cm}}\right) = 36.9°$$

Then,

$$1.00\sin\theta_2 = 1.60\sin\theta_1 = (1.60)\left(\frac{12.00}{20.0}\right) \rightarrow \theta_2 = 73.7°$$

$$f_2 = \frac{h_2}{\tan(\theta_1 - \theta_2)} = \frac{12.0 \text{ cm}}{\tan 36.8°} = 16.0 \text{ cm}$$

$$x_2 = f_2 - R(1 - \cos\theta_2)$$
$$= (16.0 \text{ cm}) - 20.0 \text{ cm}[1 - \cos(36.9°)] = 12.0 \text{ cm}$$

Now $\Delta x = x_1 - x_2 = 33.3 \text{ cm} - 12.0 \text{ cm} = \boxed{21.3 \text{ cm}}$

Section 36.6 The Camera

P36.56 The same light intensity is received from the subject, and the same light energy on the film is required:

$$IA_1\Delta t_1 = IA_2\Delta t_2$$
$$\frac{\pi d_1^2}{4}\Delta t_1 = \frac{\pi d_2^2}{4}\Delta t_2$$

Substituting f-stops and shutter speeds,

$$\left(\frac{f}{4}\right)^2\left(\frac{1}{15} \text{ s}\right) = d_2^2\left(\frac{1}{125} \text{ s}\right)$$

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solving,

$$d_2 = \sqrt{\frac{125}{15}}\frac{f}{4} = \frac{f}{1.39} = \boxed{\frac{f}{1.4}}$$

We can verify this by noting that changing the shutter speed from $\frac{1}{15\text{ s}}$ to $\frac{1}{125\text{ s}}$ is approximately a factor of 8 decrease in the exposure time, and requires a three *f*-stop increase (each increasing the area by a factor of 2), from *f*-4 down to *f*-2.8, *f*-2.0, and *f*-1.4.

***P36.57** To properly focus the image of a distant object, the lens must be at a distance equal to the focal length from the film (q_1 = 65.0 mm). For the closer object:

$$\frac{1}{p_2} + \frac{1}{q_2} = \frac{1}{f}$$

becomes $\frac{1}{2\,000\text{ mm}} + \frac{1}{q_2} = \frac{1}{65.0\text{ mm}}$,

and $q_2 = (65.0\text{ mm})\left(\frac{2\,000}{2\,000 - 65.0}\right)$.

The lens must be moved $\boxed{\text{away from the film}}$ by a distance

$$D = q_2 - q_1 = (65.0\text{ mm})\left(\frac{2\,000}{2\,000 - 65.0}\right) - 65.0\text{ mm} = \boxed{2.18\text{ mm}}$$

Section 36.7 The Eye

P36.58 The lens should take parallel light rays from a very distant object ($p = \infty$) and make them diverge from a virtual image at the woman's far point, which is 25.0 cm beyond the lens, at q = –25.0 cm.

(a) $P = \frac{1}{f} = \frac{1}{p} + \frac{1}{q} = \frac{1}{\infty} - \frac{1}{0.250\text{ m}} = \boxed{-4.00\text{ diopters}}$

(b) The power is negative: a $\boxed{\text{diverging lens}}$.

***P36.59** The corrective lens must form an upright, virtual image at the near point of the eye (i.e., q = –60.0 cm in this case) for objects located 25.0 cm in front of the eye (p = +25.0 cm). From the thin-lens equation,

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

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the required focal length of the corrective lens is

$$f = \frac{pq}{p+q} = \frac{(25.0\text{ cm})(-60.0\text{ cm})}{25.0\text{ cm} - 60.0\text{ cm}} = \boxed{+42.9\text{ cm}}$$

and the power (in diopters) of this lens will be

$$P = \frac{1}{f_{\text{in meters}}} = \frac{1}{+0.429\text{ m}} = \boxed{+2.33\text{ diopters}}$$

***P36.60** (a) $f = \frac{1}{P} = \frac{1}{-4.00\text{ diopters}} = -0.250\text{ m} = \boxed{-25.0\text{ cm}}$

(b) The corrective lens forms virtual images of very distant objects ($p \to \infty$) at $q = f = -25.0$ cm. Thus, the person must be very $\boxed{\text{nearsighted}}$, unable to see objects clearly when they are more than 25.0 cm + 2.00 cm = 27.0 cm from the eye.

(c) If contact lenses are to be worn, the far point of the eye will be 27.0 cm in front of the lens, so the needed focal length will be $f = q = -27.0$ cm, and the power is

$$P = \frac{1}{f_{\text{in meters}}} = \frac{1}{-0.270\text{ m}} = \boxed{-3.70\text{ diopters}}$$

P36.61 For starlight going through a nearsighted person's glasses,

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

$$\frac{1}{\infty} + \frac{1}{(-0.800\text{ m})} = \frac{1}{f} = -1.25\text{ diopters}$$

For a nearby object (the image is virtual),

$$\frac{1}{p} + \frac{1}{(-0.180\text{ m})} = -1.25\text{ m}^{-1}$$

so $p = \boxed{23.2\text{ cm}}$.

***P36.62** (a) When the child clearly sees objects at her far point $(p_{\max} = 125\text{ cm})$, the lens-cornea combination has assumed a focal length suitable for forming the image on the retina ($q = 2.00$ cm). The thin-lens equation gives the optical power under these conditions as

$$P_{\text{far}} = \frac{1}{f_{\text{in meters}}} = \frac{1}{p} + \frac{1}{q} = \frac{1}{1.25\text{ m}} + \frac{1}{0.020\ 0\text{ m}}$$
$$= +50.8\text{ diopters}$$

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When the eye is focused (q = 2.00 cm) on objects at her near point $\left(p_{\min} = 10.0 \text{ cm}\right)$, the optical power of the lens-cornea combination is

$$P_{\text{near}} = \frac{1}{f_{\text{in meters}}} = \frac{1}{p} + \frac{1}{q} = \frac{1}{0.100 \text{ m}} + \frac{1}{0.020\ 0 \text{ m}}$$
$$= +60.0 \text{ diopters}$$

Therefore, the range of the power of the lens-cornea combination is $\boxed{+50.8 \text{ diopters} \le P \le 60.0 \text{ diopters}}$.

(b) If the child is to see very distant objects ($p \to \infty$) clearly, her eyeglass lens must form an erect, virtual image at the far point of her eye (q = –125 cm). The optical power of the required lens is

$$P = \frac{1}{f_{\text{in meters}}} = \frac{1}{p} + \frac{1}{q} = 0 + \frac{1}{-0.125 \text{ m}} = \boxed{-0.800 \text{ diopters}}$$

Since the power, and hence the focal length, of this lens is negative, it is $\boxed{\text{diverging}}$.

***P36.63** (a) The upper portion of the lens should form an upright, virtual image of very distant objects $\left(p \approx \infty\right)$ at the far point of the eye (q = –1.50 m). The thin-lens equation then gives $f = q$ = –1.50 m, so the needed power is

$$P = \frac{1}{f_{\text{in meters}}} = \frac{1}{-1.50 \text{ m}} = \boxed{-0.667 \text{ diopters}}$$

(b) The lower part of the lens should form an upright, virtual image at the near point of the eye (q = –30.0 cm) when the object distance is p = 25.0 cm. From the thin-lens equation,

$$f = \frac{pq}{p+q} = \frac{(25.0 \text{ cm})(-30.0 \text{ cm})}{25.0 \text{ cm} - 30.0 \text{ cm}} = +1.50 \times 10^2 \text{ cm} = \boxed{+1.50 \text{ m}}$$

Therefore, the power is $P = \frac{1}{f} = \frac{1}{+1.50 \text{ m}} = \boxed{+0.667 \text{ diopters}}$.

***P36.64** $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$ so $\frac{1.00}{\infty} + \frac{1.40}{21.0 \text{ mm}} = \frac{1.40 - 1.00}{6.00 \text{ mm}}$

and 0.066 7 = 0.066 7.

They agree. $\boxed{\text{The image is inverted, real, and diminished.}}$

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***P36.65** (a) Yes, a single lens can correct the patient's vision. The patient needs corrective action in both the near vision (to allow clear viewing of objects between 45.0 cm and the normal near point of 25.0 cm) and the distant vision (to allow clear viewing of objects more than 85.0 cm away). A single lens solution is for the patient to wear a bifocal or progressive lens. Alternately, the patient must purchase two pairs of glasses, one for reading, and one for distant vision.

(b) To correct the near vision, the lens must form an upright, virtual image at the patient's near point ($q = -45.0$ cm) when a real object is at the normal near point ($p = +25.0$ cm). The thin-lens equation gives the needed focal length as

$$f = \frac{pq}{p+q} = \frac{(25.0\text{ cm})(-45.0\text{ cm})}{25.0\text{ cm} - 45.0\text{ cm}} = \boxed{+56.3\text{ cm}}$$

so the required power in diopters is

$$P = \frac{1}{f_{\text{in meters}}} = \frac{1}{+0.563\text{ m}} = \boxed{+1.78\text{ diopters}}$$

(c) To correct the distant vision, the lens must form an upright, virtual image at the patient's far point ($q = -85.0$ cm) for the most distant objects ($p \to \infty$). The thin-lens equation gives the needed focal length as $f = q = -85.0$ cm, so the needed power is

$$P = \frac{1}{f_{\text{in meters}}} = \frac{1}{-0.850\text{ m}} = \boxed{-1.18\text{ diopters}}$$

Section 36.8 The Simple Magnifyer

P36.66 (a) Angular magnification is a maximum when the image is at the near point of the eye: $q = 25.0$ cm. From the thin lens equation:

$$\frac{1}{p} + \frac{1}{(-25.0\text{ cm})} = \frac{1}{5.00\text{ cm}} \quad \text{or} \quad p = \boxed{4.17\text{ cm}}$$

(b) From Equation 36.24,

$$M = -\frac{q}{p} = 1 + \frac{25.0\text{ cm}}{f} = 1 + \frac{25.0\text{ cm}}{5.00\text{ cm}} = \boxed{6.00}$$

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Section 36.9 The Compound Microscope

P36.67 Using Equation 36.26,

$$M \approx -\left(\frac{L}{f_o}\right)\left(\frac{25.0\text{ cm}}{f_e}\right) = -\left(\frac{23.0\text{ cm}}{0.400\text{ cm}}\right)\left(\frac{25.0\text{ cm}}{2.50\text{ cm}}\right) = \boxed{-575}$$

Section 36.10 The Telescope

P36.68 $f_o = 20.0$ m, $f_e = 0.025\,0$ m

(a) From Equation 36.27, The angular magnification produced by this telescope is

$$m = -\frac{f_o}{f_e} = \boxed{-800}$$

(b) Since $m < 0$, the image is $\boxed{\text{inverted}}$.

P36.69 Let I_0 represent the intensity of the light from the nebula and θ_0 its angular diameter. With the first telescope, the image diameter h' on the film is given by

$$\theta_o = -\frac{h'}{f_o} \text{ as } h' = -\theta_o(2\,000\text{ mm})$$

The light power captured by the telescope aperture is

$$P_1 = I_0 A_1 = I_0\left[\frac{\pi(200\text{ mm})^2}{4}\right]$$

and the light energy focused on the film during the exposure is

$$E_1 = P_1 \Delta t_1 = I_0\left[\frac{\pi(200\text{ mm})^2}{4}\right](1.50\text{ min})$$

Likewise, the light power captured by the aperture of the second telescope is

$$P_2 = I_0 A_2 = I_0\left[\frac{\pi(60.0\text{ mm})^2}{4}\right]$$

and the light energy is

$$E_2 = I_0\left[\frac{\pi(60.0\text{ mm})^2}{4}\right]\Delta t_2$$

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Therefore, to have the same light energy per unit area, it is necessary that

$$\frac{I_0\left[\pi(60.0\text{ mm})^2/4\right]\Delta t_2}{\pi\left[\theta_o(900\text{ mm})^2/4\right]} = \frac{I_0\left[\pi(200\text{ mm})^2/4\right](1.50\text{ min})}{\pi\left[\theta_o(2\,000\text{ mm})^2/4\right]}$$

The required exposure time with the second telescope is

$$\Delta t_2 = \frac{(200\text{ mm})^2(900\text{ mm})^2}{(60.0\text{ mm})^2(2\,000\text{ mm})^2}(1.50\text{ min}) = \boxed{3.38\text{ min}}$$

P36.70 (a) The mirror-and-lens equation, $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$, gives

$$q = \frac{1}{1/f - 1/p} = \frac{1}{(p-f)/fp} = \frac{fp}{p-f}$$

Then,

$$M = \frac{h'}{h} = -\frac{q}{p} = -\frac{f}{p-f}$$

gives $h' = \dfrac{fh}{f-p}$

(b) For $p >> f$, $f - p \approx -p$. Then, $h' = \boxed{-\dfrac{hf}{p}}$

(c) Suppose the telescope observes the space station at the zenith:

$$h' = -\frac{hf}{p} = -\frac{(108.6\text{ m})(4.00\text{ m})}{407\times10^3\text{ m}} = \boxed{-1.07\text{ mm}}$$

Additional Problems

P36.71 (a) For the lens in air,

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\frac{1}{79.0\text{ cm}} = (1.55-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

For the same lens in water,

$$\frac{1}{f'} = \left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

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Substituting,

$$\frac{1}{f'} = \left(\frac{1.55}{1.333} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

By division,

$$\frac{1/79.0\text{ cm}}{1/f'} = \frac{0.55}{\left(\frac{1.55}{1.333} - 1\right)} = \frac{f'}{79.0\text{ cm}} \rightarrow f' = \boxed{267\text{ cm}}$$

(b) The path of a reflected ray does not depend on the refractive index of the medium which the reflecting surface bounds. Therefore the focal length of a mirror does not change when it is put into a different medium: $f' = \frac{R}{2} = f = \boxed{79.0\text{ cm}}$.

P36.72 The real image formed by the concave mirror serves as a real object for the convex mirror with $p = 50$ cm and $q = -10$ cm. Therefore,

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q} \quad \rightarrow \quad \frac{1}{f} = \frac{1}{50.0\text{ cm}} + \frac{1}{(-10.0\text{ cm})}$$

gives $f = -12.5$ cm and $R = 2f = \boxed{-25.0\text{ cm}}$.

P36.73 Only a diverging lens gives an upright, diminished image. Therefore, the image is virtual and between the object and the lens (the image is closer to the lens), and $q < 0$. We have

$$d = p - |q| = p + q, \quad \text{and} \quad M = -\frac{q}{p},$$

so $q = -Mp$ and $d = p - Mp$.

Therefore, $p = \frac{d}{1-M}$:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \frac{1}{p} + \frac{1}{-Mp} = \frac{-M+1}{-Mp} = \frac{(1-M)^2}{-Md}$$

$$f = \frac{-Md}{(1-M)^2} = \frac{-(0.500)(20.0\text{ cm})}{(1-0.500)^2} = \boxed{-40.0\text{ cm}}$$

P36.74 For a single lens, an object and its image cannot be on opposite sides of the lens if the image is upright. The object and image must be on the same side of the lens; thus the image is virtual, and $q < 0$. Because the image is upright, $M > 0$.

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If the image is between the object and the lens (the image is closer to the lens), we have

$$d = p - |q| = p + q, \quad \text{so} \quad q = d - p:$$

$$M = -\frac{q}{p} \quad \text{so} \quad q = -Mp = d - p \;\rightarrow\; p = \frac{d}{1-M}$$

Substituting into the thin lens equation, $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$, gives

$$\frac{1}{p} + \frac{1}{(-Mp)} = \frac{1}{f}$$

Solving,

$$\frac{M}{Mp} + \frac{1}{(-Mp)} = \frac{1}{f} = \frac{M-1}{Mp} = \frac{M-1}{M}\left(\frac{1-M}{d}\right) = -\frac{(1-M)^2}{Md}$$

$$\rightarrow \boxed{f = \frac{-Md}{(1-M)^2}}$$

Since M is positive, the lens is diverging.

If the object is between the image and the lens (the object is closer to the lens), the lens is converging. We have

$$d = |q| - p = -q - p \quad \rightarrow \quad q = -d - p$$

Substituting into the thin lens equation, $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$, gives

$$\frac{1}{p} + \frac{1}{(-Mp)} = \frac{1}{f}$$

Solving,

$$\frac{M}{Mp} + \frac{1}{(-Mp)} = \frac{1}{f} = \frac{M-1}{Mp} = \frac{M-1}{M}\left(\frac{M-1}{d}\right) = \frac{(M-1)^2}{Md}$$

$$\rightarrow \boxed{f = \frac{Md}{(M-1)^2}}$$

Since M is positive, the lens is converging.

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***P36.75** The lens for the left eye forms an upright, virtual image at q_L = –50.0 cm when the object distance is p_L = 25.0 cm, so the thin lens equation $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$, gives its focal length as

$$f_L=\frac{p_L q_L}{p_L+q_L}=\frac{(25.0\text{ cm})(-50.0\text{ cm})}{25.0\text{ cm}-50.0\text{ cm}}=50.0\text{ cm}$$

Similarly for the other lens, q_R = –100 cm when p_R = 25.0 cm, and f_R = 33.3 cm.

(a) Using the lens for the left eye as the objective,

$$m=\frac{f_o}{f_c}=\frac{50.0\text{ cm}}{33.3\text{ cm}}=\boxed{1.50}$$

(b) Using the lens for the right eye as the eyepiece and, for maximum magnification, requiring that the final image be formed at the normal near point (q_e =–25.0 cm) gives the object distance for the eyepiece as

$$p_e=\frac{q_e f_e}{q_e-f_e}=\frac{(-25.0\text{ cm})(33.3\text{ cm})}{-25.0\text{ cm}-33.3\text{ cm}}=+14.3\text{ cm}$$

The maximum magnification by the eyepiece is then

$$m_e=1+\frac{25.0\text{ cm}}{f_e}=1+\frac{25.0\text{ cm}}{33.3\text{ cm}}=+1.75$$

and the image distance for the objective is

$$q_1=L-p_e=10.0\text{ cm}-14.3\text{ cm}=-4.28\text{ cm}$$

The thin lens equation then gives the object distance for the objective as

$$p_1=\frac{q_1 f_1}{q_1-f_1}=\frac{(-4.28\text{ cm})(50.0\text{ cm})}{-4.28\text{ cm}-50.0\text{ cm}}=+3.95\text{ cm}$$

The magnification by the objective is then

$$M_1=-\frac{q_1}{p_1}=-\frac{(-4.28\text{ cm})}{3.95\text{ cm}}=+1.08$$

and the overall magnification is

$$m=M_1 m_e=(+1.08)(+1.75)=\boxed{1.90}$$

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***P36.76** The image will be inverted. With $h = 6.00$ cm, we require $h' = -1.00$ mm.

(a) $M = \dfrac{h'}{h} = -\dfrac{q}{p}$ gives

$$q = -p\frac{h'}{h} = -(50.0\text{ mm})\left(\frac{-1.00\text{ mm}}{60.0\text{ mm}}\right) = \boxed{0.833\text{ mm}}$$

(b) From $\dfrac{1}{p} + \dfrac{1}{q} = \dfrac{1}{f} = \dfrac{1}{50.0\text{ mm}} + \dfrac{1}{0.833\text{ mm}}$, we obtain

$$f = \boxed{0.820\text{ mm}}$$

P36.77 (a) Start with the first pass through the lens.

$$\frac{1}{q_1} = \frac{1}{f_1} - \frac{1}{p_1} = \frac{1}{80.0\text{ cm}} - \frac{1}{100\text{ cm}}$$

gives $q_1 = +400$ cm or 400 cm to right of the lens.

The object of the mirror is 400 cm – 100 cm = 300 cm to the right of the mirror, so the object is virtual. Therefore, for the mirror, $p_2 = -300$ cm:

$$\frac{1}{q_2} = \frac{1}{f_2} - \frac{1}{p_2} = \frac{1}{(-50.0\text{ cm})} - \frac{1}{(-300\text{ cm})}$$

gives $q_2 = -60.0$ cm or 60.0 cm to the right of the mirror.

The image formed by the mirror is 100 cm + 60 cm = 160 cm to the right of the lens. Therefore, for the second pass through the lens, $p_3 = 160$ cm:

$$\frac{1}{q_3} = \frac{1}{f_1} - \frac{1}{p_3} = \frac{1}{80.0\text{ cm}} - \frac{1}{160\text{ cm}}$$

or $q_3 = \boxed{160\text{ cm to the left of lens}}$.

(b) $M_1 = -\dfrac{q_1}{p_1} = -\dfrac{400\text{ cm}}{100\text{ cm}} = -4.00 \qquad M_2 = -\dfrac{q_2}{p_2} = -\dfrac{(-60.0\text{ cm})}{(-300\text{ cm})} = -\dfrac{1}{5}$

$$M_3 = -\frac{q_3}{p_3} = -\frac{160\text{ cm}}{160\text{ cm}} = -1 \qquad M = M_1M_2M_3 = \boxed{-0.800}$$

(c) Since $M < 0$ the final image is $\boxed{\text{inverted}}$.

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P36.78 (a) We start with the final image and work backward. From Figure P36.78, the final image is virtual (to left of lens 2) and x = 30.0 cm, so

$$q_2 = -(50.0\text{ cm} - 30.0\text{ cm}) = -20.0\text{ cm}$$

The thin lens equation then gives

$$\frac{1}{p_2} + \frac{1}{q_2} = \frac{1}{f_2}: \quad \frac{1}{p_2} + \frac{1}{-20.0\text{ cm}} = \frac{1}{20.0\text{ cm}} \rightarrow \quad p_2 = +10.0\text{ cm}$$

The image formed by the first lens serves as the object for the second lens and is located 10.0 cm in front of the second lens.

Thus, q_1 = 50.0 cm – 10.0 cm = 40.0 cm and the thin lens equation gives

$$\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{f_1}: \quad \frac{1}{p_1} + \frac{1}{40.0\text{ cm}} = \frac{1}{10.0\text{ cm}} \rightarrow \quad p_1 = +13.3\text{ cm}$$

The original object should be located $\boxed{\text{13.3 cm in front of the first lens}}$.

(b) The overall magnification is

$$M = M_1 M_2 = \left(-\frac{q_1}{p_1}\right)\left(-\frac{q_2}{p_2}\right) = \left(-\frac{40.0\text{ cm}}{13.3\text{ cm}}\right)\left(-\frac{(-20.0\text{ cm})}{10.0\text{ cm}}\right)$$

$$= \boxed{-6.00}$$

(c) Since $M < 0$, the final image is $\boxed{\text{inverted}}$.

(d) Since $q_2 < 0$, it is $\boxed{\text{virtual}}$.

P36.79 (a) With light going through the piece of glass from left to right, the radius of the first surface is positive and that of the second surface is negative according to the sign convention of Table 36.2. Thus, $R_1 = +2.00$ cm and $R_2 = -4.00$ cm. Applying $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$ to the first surface gives

$$\frac{1.00}{1.00\text{ cm}} + \frac{1.50}{q_1} = \frac{1.50 - 1.00}{+2.00\text{ cm}}$$

which yields $q_1 = -2.00$ cm. The first surface forms a virtual image 2.00 cm to the left of that surface and 16.0 cm to the left of the second surface.

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The image formed by the first surface is the object for the second surface, so $p_2 = +16.0\text{ cm}$ and $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$ gives

$$\frac{1.50}{16.0\text{ cm}} + \frac{1.00}{q_2} = \frac{1.00 - 1.50}{-4.00\text{ cm}} \quad \text{or} \quad q_2 = +32.0\text{ cm}$$

32.0 cm to the right of the second surface

(b) The final image distance is positive, so the image is real.

P36.80 (a) When the meterstick coordinate of the object is 0, its object distance is $p_i = 32$ cm. When the meterstick coordinate of the object is x, its object distance is $p = 32\text{ cm} - x$. The image distance from the lens is given by the thin lens equation, $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$ (in the following, all variables are in units of cm, and units are suppressed). Substituting,

$$\frac{1}{32.0 - x} + \frac{1}{q} = \frac{1}{26.0}$$

Solving for q then gives

$$\frac{1}{q} = \frac{1}{26.0} - \frac{1}{(32.0 - x)} = \frac{(32.0 - x) - 26.0}{26.0(32.0 - x)} = \frac{6.0 - x}{26.0(32.0 - x)}$$

$$q = \frac{832 - 26.0x}{6.0 - x}$$

The image distance q is measured from the position of the lens. The image coordinate on the meterstick is

$$x' = 32.0 + q = 32.0 + \frac{832 - 26.0x}{6.0 - x} = \frac{32.0(6.0 - x) + 832 - 26.0x}{6.0 - x}$$

$$x' = \frac{1024 - 58.0x}{6.0 - x} \text{ where } x \text{ and } x' \text{ are in centimeters.}$$

(b) The image starts at the position $x_i' = 171$ cm and moves in the positive x direction, faster and faster, and as the object approaches the position $x = 6$ cm (the focal point of the lens), the image goes out to infinity. At the instant the object is at $x = 6$ cm, the rays from the top of the object are parallel as they leave the lens: their intersection point can be described as at $x' = \infty$ to the right or equally well at $x' = -\infty$ on the left. From $x' = -\infty$ the image continues moving to the right, now slowing down. It reaches, for example, –280 cm when the object is at 8 cm, and –55 cm when the object is finally at 12 cm.

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object position (cm)	image position (cm)
x	x'
0	170.7
1	193.2
2	227.0
3	283.3
4	396.0
5	734.0
6	infinity
7	–618.0
8	–280.0
9	–167.3
10	–111.0
11	–77.2
12	–54.7

(c) The image moves to infinity and beyond—meaning it moves forward to infinity (on the right), jumps back to minus infinity (on the left), and then proceeds forward again.

(d) The image usually travels to the right, except when it jumps from plus infinity (right) to minus infinity (left).

P36.81 (a) $$\frac{1}{p_1}+\frac{1}{q_1}=\frac{1}{f_1}=\frac{2}{R} \rightarrow \frac{1}{q_1}=\frac{1}{f_1}-\frac{1}{p_1}=\frac{1}{10.0\text{ cm}}-\frac{1}{12.5\text{ cm}}$$

so $q_1 = 50.0$ cm (a real image, to left of mirror).This serves as an object for the lens (a virtual object, to left of lens) with object distance $p_2 = 25.0\text{ cm} - 50.0\text{ cm} = -25.0$ cm, so

$$\frac{1}{q_2}=\frac{1}{f_2}-\frac{1}{p_2}=\frac{1}{(-16.7\text{ cm})}-\frac{1}{(-25.0\text{ cm})}$$

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so $q_2 = -50.3$ cm (a virtual image),

meaning 50.3 cm to the right of the lens. Thus, the final image is located $\boxed{25.3 \text{ cm to right of mirror}}$.

(b) The final image distance is negative (–50.3 cm), so the image is $\boxed{\text{virtual}}$.

Calculate the overall magnification $M = M_1 M_2$:

$$M_1 = -\frac{q_1}{p_1} = -\frac{50.0 \text{ cm}}{12.5 \text{ cm}} = -4.00$$

$$M_2 = -\frac{q_2}{p_2} = -\frac{(-50.3 \text{ cm})}{(-25.0 \text{ cm})} = -2.01$$

Then $M = M_1 M_2 = 8.05$.

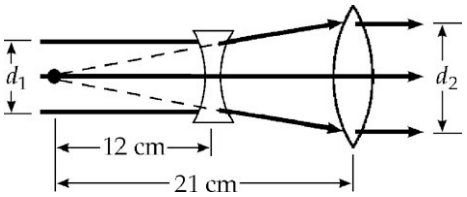
(c) The magnification is positive, so the image is $\boxed{\text{upright}}$.

(d) From above, $M = M_1 M_2 = \boxed{8.05}$.

P36.82 (a) Have the beam pass through the diverging lens first, then the converging lens. The rays of light entering the diverging lens are parallel, so they behave as though they come from an object at infinity ($p = \infty$):

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \qquad \frac{1}{\infty} + \frac{1}{q} = \frac{1}{-12.0 \text{ cm}}$$

or $q = -12.0$ cm.



ANS. FIG. P36.82

Use this image as a real object for the converging lens, placing it at the focal point on the object side of the lens, at $p = 21.0$ cm. Then

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \quad \rightarrow \quad \frac{1}{21 \text{ cm}} + \frac{1}{q} = \frac{1}{21 \text{ cm}}$$

or $q = \infty$.

The exiting rays will be parallel. The lenses must be 21.0 cm – 12.0 cm = 9.00 cm apart.

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(b) Refer to ANS. FIG. P36.82. By similar triangles,

$$\frac{d_2}{d_1} = \frac{21.0 \text{ cm}}{12.0 \text{ cm}} = \boxed{1.75 \text{ times}}$$

P36.83 (a) $$I = \frac{P}{4\pi r^2} = \frac{4.50 \text{ W}}{4\pi\left(1.60 \times 10^{-2} \text{ m}\right)^2} = \boxed{1.40 \text{ kW/m}^2}$$

(b) $$I = \frac{P}{4\pi r^2} = \frac{4.50 \text{ W}}{4\pi(7.20 \text{ m})^2} = \boxed{6.91 \text{ mW/m}^2}$$

(c) $$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} : \frac{1}{7.20 \text{ m}} + \frac{1}{q} = \frac{1}{0.350 \text{ m}}$$

$$\rightarrow \quad q = 0.368 \text{ m}$$

$$M = \frac{h'}{3.20 \text{ cm}} = -\frac{q}{p} = -\frac{0.368 \text{ m}}{7.20 \text{ m}}$$

$$\rightarrow \quad h' = \boxed{0.164 \text{ cm}}$$

(d) The lens intercepts power given by

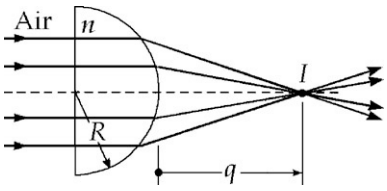
$$P = IA = \left(6.91 \times 10^{-3} \text{ W/m}^2\right)\left[\frac{\pi}{4}(0.150 \text{ m})^2\right]$$

and puts it all onto the image where

$$I = \frac{P}{A} = \frac{\left(6.91 \times 10^{-3} \text{ W/m}^2\right)\left[\pi(15.0 \text{ cm})^2/4\right]}{\pi(0.164 \text{ cm})^2/4}$$

$$I = \boxed{58.1 \text{ W/m}^2}$$

P36.84 A hemisphere is too thick to be described as a thin lens. The light is undeviated on entry into the flat face. We next consider the light's exit from the second surface, for which $R = -6.00$ cm. The incident rays are parallel, so $p = \infty$.



ANS. FIG. P36.84

Then, $$\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$$

becomes $$0 + \frac{1}{q} = \frac{1.00 - 1.56}{-6.00 \text{ cm}}$$

and $\boxed{q = 10.7 \text{ cm}}$.

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P36.85 Use the lens makers' equation, Equation 36.15, and the conventions of Table 36.2. The first lens has focal length described by

$$\frac{1}{f_1} = (n_1 - 1)\left(\frac{1}{R_{1,1}} - \frac{1}{R_{1,2}}\right) = (n_1 - 1)\left(\frac{1}{\infty} - \frac{1}{R}\right) = \frac{1 - n_1}{R}$$

For the second lens

$$\frac{1}{f_2} = (n_2 - 1)\left(\frac{1}{R_{2,1}} - \frac{1}{R_{2,2}}\right) = (n_2 - 1)\left(\frac{1}{+R} - \frac{1}{-R}\right) = +\frac{2(n_2 - 1)}{R}$$

Let an object be placed at any distance p_1 large compared to the thickness of the doublet. The first lens forms an image according to

$$\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{f_1}$$

$$\frac{1}{q_1} = \frac{1 - n_1}{R} - \frac{1}{p_1}$$

This virtual ($q_1 < 0$) image (to the left of lens 1) is a real object for the second lens at distance $p_2 = -q_1$. For the second lens

$$\frac{1}{p_2} + \frac{1}{q_2} = \frac{1}{f_2}$$

$$\frac{1}{q_2} = \frac{2n_2 - 2}{R} - \frac{1}{p_2} = \frac{2n_2 - 2}{R} + \frac{1}{q_1} = \frac{2n_2 - 2}{R} + \frac{1 - n_1}{R} - \frac{1}{p_1}$$

$$= \frac{2n_2 - n_1 - 1}{R} - \frac{1}{p_1}$$

Then $\frac{1}{p_1} + \frac{1}{q_2} = \frac{2n_2 - n_1 - 1}{R}$ so the doublet behaves like a single lens with $\frac{1}{f} = \frac{2n_2 - n_1 - 1}{R}$.

P36.86 Find the image position for light traveling to the left through the lens:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f_L} \rightarrow q = \frac{pf_L}{p - f_L} = \frac{(0.300\ \text{m})(0.200\ \text{m})}{0.300\ \text{m} - 0.200\ \text{m}} = 0.600\ \text{m}$$

Therefore, this image forms 0.600 m to the left of the lens. Find the image formed by light traveling to the right toward the mirror from an object distance of 1.30 m – 0.300 m = 1.00 m:

$$\frac{1}{p_M} + \frac{1}{q_M} = \frac{1}{f_M}$$

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Solving and substituting numerical values gives

$$q_M = \frac{p_M f_M}{p_M - f_M} = \frac{(1.00\ \text{m})(0.500\ \text{m})}{1.00\ \text{m} - 0.500\ \text{m}} = 1.00\ \text{m}$$

This image forms at the position of the original object. Therefore, as light continues to the left through the lens, it will form an image at a position 0.600 m to the left of the lens. As a result, *both* images form at the *same* position and there are not two locations at which the student can hold a screen to see images formed by this system.

P36.87 For the first lens, the thin lens equation gives

$$q_1 = \frac{f_1 p_1}{p_1 - f_1} = \frac{(-6.00\ \text{cm})(12.0\ \text{cm})}{12.0\ \text{cm} - (-6.00\ \text{cm})} = -4.00\ \text{cm}$$

The first lens forms an image 4.00 cm to its left. The rays between the lenses diverge from this image, so the second lens receives diverging light. It sees a real object at distance

$$p_2 = d - (-4.00\ \text{cm}) = d + 4.00\ \text{cm}$$

For the second lens, when we require that $q_2 \to \infty$, the mirror-lens equation becomes $p_2 = f_2 = 12.0$ cm.

Since the object for the converging lens must be 12.0 cm to its left, and since this object is the image for the diverging lens, which is 4.00 cm to **its** left, the two lenses must be separated by 8.00 cm.

Mathematically,

$$d + 4.00\ \text{cm} = f_2 = 12.0\ \text{cm} \ \rightarrow \ d = \boxed{8.00\ \text{cm}}$$

P36.88 For the first lens, the thin lens equation gives

$$q_1 = \frac{f_1 p}{p - f_1}$$

We require that $q_2 \to \infty$ for the second lens; the thin lens equation gives $p_2 = f_2$, where, in this case,

$$p_2 = d - q_1 = d - \frac{f_1 p}{p - f_1}$$

Therefore, from $p_2 = f_2$,

$$d - \frac{f_1 p}{p - f_1} = f_2$$

$$d = \frac{f_1 p}{p - f_1} + f_2 = \frac{f_1 p + f_2 (p - f_1)}{p - f_1} = \boxed{\frac{p(f_1 + f_2) - f_1 f_2}{p - f_1}}$$

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P36.89 The inverted image is formed by light that leaves the object and goes directly through the lens, never having reflected from the mirror. For the formation of this inverted image, we have

$$M = -\frac{q_1}{p_1} = -1.50 \quad \text{giving} \quad q_1 = +1.50p_1$$

The thin lens equation then gives (with p and q in centimeters)

$$\frac{1}{p_1} + \frac{1}{1.50p_1} = \frac{1}{10.0}$$

$$\frac{1.50}{1.50p_1} + \frac{1}{1.50p_1} = \frac{1}{10.0}$$

$$\frac{2.50}{1.50p_1} = \frac{1}{10.0}$$

giving $$p_1 = 10.0\left(\frac{2.50}{1.50}\right) = \boxed{16.7 \text{ cm}}.$$

The upright image is formed by light that passes through the lens after reflecting from the mirror. The object for the lens in this upright image formation is the image formed by the mirror. In order for the lens to form the upright image at the same location as the inverted image, the image formed by the mirror must be located at the position of the original object (so the object distances, and hence image distances, are the same for both the inverted and upright images formed by the lens). Therefore, the object distance and the image distance for the mirror are equal, and their common value is

$$q_{\text{mirror}} = p_{\text{mirror}} = 40.0 - p_1 = 40.0 - 16.7 = +23.3$$

The mirror equation, $\frac{1}{p_{\text{mirror}}} + \frac{1}{q_{\text{mirror}}} = \frac{1}{f_{\text{mirror}}}$, then gives

$$\frac{1}{f_{\text{mirror}}} = \frac{1}{23.3 \text{ cm}} + \frac{1}{23.3 \text{ cm}} = \frac{2}{23.3 \text{ cm}}$$

or $$f_{\text{mirror}} = +\frac{23.3 \text{ cm}}{2} = \boxed{+11.7 \text{ cm}}.$$

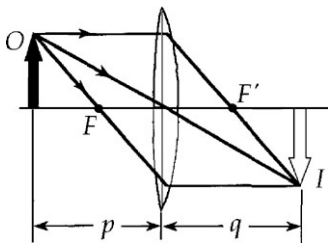
P36.90 (a) In the first situation, $\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{f}$, and

$$p_1 + q_1 = 1.50 \;\rightarrow\; q_1 = 1.50 - p_1$$

where f, p, and q are in meters.

Substituting, we have $\boxed{\frac{1}{f} = \frac{1}{p_1} + \frac{1}{1.50 - p_1}}$.

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ANS. FIG. P36.90

(b) In the second situation, $\frac{1}{p_2}+\frac{1}{q_2}=\frac{1}{f}$,

$$p_2 = p_1 + 0.900 \text{ m} \quad \text{and} \quad q_2 = q_1 - 0.900 \text{ m} = 0.600 \text{ m} - p_1,$$

where f, p, and q are in meters.

Substituting, we have $\boxed{\frac{1}{f}=\frac{1}{p_1+0.900}+\frac{1}{0.600-p_1}}$.

(c) Both lens equation are equal:

$$\frac{1}{p_1}+\frac{1}{q_1}=\frac{1}{f}=\frac{1}{p_2}+\frac{1}{q_2}$$

$$\frac{1}{p_1}+\frac{1}{1.50-p_1}=\frac{1}{p_1+0.900}+\frac{1}{0.600-p_1}$$

$$\frac{1.50-p_1+p_1}{p_1(1.50-p_1)}=\frac{0.600-p_1+p_1+0.900}{(p_1+0.900)(0.600-p_1)}$$

$$\frac{1.50}{p_1(1.50-p_1)}=\frac{1.50}{(p_1+0.900)(0.600-p_1)}$$

Simplified, this becomes

$$p_1(1.50-p_1)=(p_1+0.900)(0.600-p_1)$$

$$1.50p_1-p_1^2=(0.600-0.900)p_1+(0.900)(0.600)-p_1^2$$

$$1.80p_1=0.540$$

$$p_1=\boxed{0.300 \text{ m}}$$

(d) From part (a), $\frac{1}{f}=\frac{1}{p_1}+\frac{1}{1.50-p_1}$:

$$\frac{1}{f}=\frac{1}{0.300}+\frac{1}{1.50-0.300}$$

$$f=\boxed{0.240 \text{ m}}$$

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P36.91 (a) For the mirror, $f = \frac{R}{2} = +1.50 \text{ m}$. In addition, because the distance to the Sun is so much larger than any other distances, we can take $p = \infty$.

The mirror equation, $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$, then gives $q = f =$ 1.50 m in front of the mirror.

(b) Now, in $M = -\frac{q}{p} = \frac{h'}{h}$,

the magnification is nearly zero, but we can be more precise: $\frac{h}{p} = 0.533°$ is the angular diameter of the object. Thus,

$$h' = -\frac{h}{p}q = -\left[(0.533°)\left(\frac{\pi \text{ rad}}{180°}\right)\right](1.50 \text{ m}) = -0.014\,0 \text{ m}$$
$$= -1.40 \text{ cm}$$

and the image diameter is 1.40 cm.

P36.92 (a) For lens one, as shown in the top panel in ANS. FIG. P36.92,

$$\frac{1}{40.0 \text{ cm}} + \frac{1}{q_1} = \frac{1}{30.0 \text{ cm}}$$
$$q_1 = 120 \text{ cm}$$

This real image is the object of the second lens: $I_1 = O_2$; it is *behind* the lens, as shown in the middle panel in ANS. FIG. P36.92, so it is a virtual object for the second lens. That is, the object distance is

$$p_2 = 110 \text{ cm} - 120 \text{ cm} = -10.0 \text{ cm}$$

$$\frac{1}{-10.0 \text{ cm}} + \frac{1}{q_2} = \frac{1}{-20.0 \text{ cm}}:$$

$q_2 =$ 20.0 cm

(b) From part (a),

$$M_1 = -\frac{q_1}{p_1} = -\frac{120 \text{ cm}}{40.0 \text{ cm}} = -3.00$$

$$M_2 = -\frac{q_2}{p_2} = -\frac{20.0 \text{ cm}}{(-10.0 \text{ cm})} = +2.00$$

$M_{\text{overall}} = M_1 M_2 =$ −6.00

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(c) $M_{\text{overall}} < 0$, so final image is $\boxed{\text{inverted}}$.

(d) If lens two is a converging lens (bottom panel in ANS. FIG. P36.92):

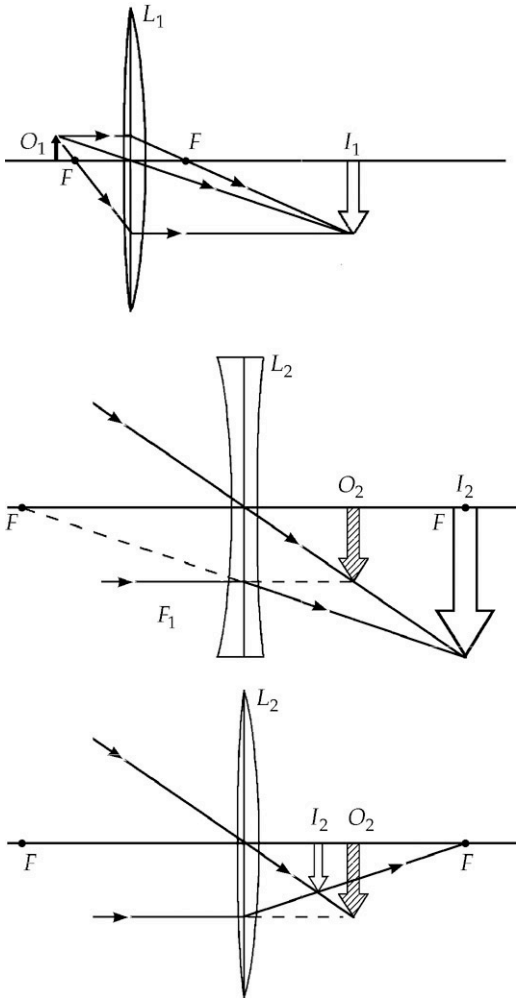
$$\frac{1}{-10.0\text{ cm}} + \frac{1}{q_2} = \frac{1}{20.0\text{ cm}}$$

$$q_2 = \boxed{6.67\text{ cm}}$$

$$M_2 = -\frac{6.67\text{ cm}}{(-10.0\text{ cm})} = +0.667$$

$$M_{\text{overall}} = M_1 M_2 = \boxed{-2.00}$$

Again, $M_{\text{overall}} < 0$ and the final image is $\boxed{\text{inverted}}$.



ANS. FIG. P36.92

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Challenge Problems

P36.93 (a) For the light the mirror intercepts, the power is given by

$$P = I_0 A = I_0 \pi R_a^2$$

Substituting,

$$350\ \text{W} = \left(1\,000\ \text{W/m}^2\right)\pi R_a^2$$

and $R_a = \boxed{0.334\ \text{m or larger}}$.

(b) In $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}=\frac{2}{R}$ we have $p \to \infty$, so $q = \frac{R}{2}$ and

$$M = \frac{h'}{h} = -\frac{q}{p},$$

so $$h' = -q\left(\frac{h}{p}\right) = -\left(\frac{R}{2}\right)\left[0.533°\left(\frac{\pi\ \text{rad}}{180°}\right)\right] = -\left(\frac{R}{2}\right)(9.30\ \text{m rad})$$

where $\frac{h}{p}$ is the angle the Sun subtends.

The intensity at the image is then

$$I = \frac{P}{\pi h'^2/4} = \frac{4I_0\pi R_a^2}{\pi h'^2} = \frac{4I_0 R_a^2}{h'^2}$$

$$I = \frac{4I_0 R_a^2}{\left(R/2\right)^2\left(9.30\times10^{-3}\ \text{rad}\right)^2}$$

$$120\times10^3\ \text{W/m}^2 = \frac{16\left(1\,000\ \text{W/m}^2\right)R_a^2}{R^2\left(9.30\times10^{-3}\ \text{rad}\right)^2}$$

$$\frac{R_a^2}{R^2} = 6.49\times10^{-4}$$

So, $\boxed{\frac{R_a}{R} = 0.025\,5\ \text{or larger}}$.

P36.94 (a) From the thin lens equation,

$$\frac{1}{q_1} = \frac{1}{f_1} - \frac{1}{p_1} = \frac{1}{5\ \text{cm}} - \frac{1}{7.5\ \text{cm}} \ \rightarrow\ q_1 = 15\ \text{cm}$$

and, from the definition of magnification,

$$M_1 = -\frac{q_1}{p_1} = -\frac{15\ \text{cm}}{7.5\ \text{cm}} = -2$$

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Then, for a combination of two lenses,

$$M = M_1 M_2 : \; 1 = (-2) M_2$$

or

$$M_2 = -\frac{1}{2} = -\frac{q_2}{p_2} \;\rightarrow\; p_2 = 2q_2$$

From the thin lens equation for the second lens,

$$\frac{1}{p_2} + \frac{1}{q_2} = \frac{1}{f_2} : \; \frac{1}{2q_2} + \frac{1}{q_2} = \frac{1}{10 \text{ cm}} \;\rightarrow\; q_2 = 15 \text{ cm}, \; p_2 = 30 \text{ cm}$$

So the distance between the object and the screen is

$$p_1 + q_1 + p_2 + q_2 = 7.5 \text{ cm} + 15 \text{ cm} + 30 \text{ cm} + 15 \text{ cm} = \boxed{67.5 \text{ cm}}$$

(b) In the following, if no units are shown, assume all distances (p, q, and f) are in units of cm.

For lens 1, we have $\frac{1}{p_1'} + \frac{1}{q_1'} = \frac{1}{f_1} = \frac{1}{5}$. Solve for q_1' in terms of p_1':

$$q_1' = \frac{5p_1'}{p_1' - 5} \qquad \textbf{[1]}$$

Now we have $M_1' = -\frac{q_1'}{p_1'} = -\frac{5}{p_1' - 5}$, using [1]. From $M' = M_1' M_2' = 3$, we have

$$M_2' = \frac{M'}{M_1'} = -\frac{3}{5}(p_1' - 5) = -\frac{q_2'}{p_2'}$$

$$q_2' = \frac{3}{5} p_2' (p_1' - 5) \qquad \textbf{[2]}$$

Substitute [2] into the lens equation for lens 2, $\frac{1}{p_2'} + \frac{1}{q_2'} = \frac{1}{f_2} = \frac{1}{10 \text{ cm}}$, and obtain p_2' in terms of p_1':

$$p_2' = \frac{10(3p_1' - 10)}{3(p_1' - 5)} \qquad \textbf{[3]}$$

Substitute [3] into [2], to obtain q_2' in terms of p_1':

$$q_2' = 2(3p_1' - 10) \qquad \textbf{[4]}$$

We know that the distance from object to the screen is a constant:

$$p_1' + q_1' + p_2' + q_2' = \text{ a constant} \qquad \textbf{[5]}$$

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Using [1], [3], and [4], and the value obtained in part (a), [5] becomes

$$p_1' + \frac{5p_1'}{p_1' - 5} + \frac{10(3p_1' - 10)}{3(p' - 5)} + 2(3p_1' - 10) = 67.5 \qquad \textbf{[6]}$$

Multiplying equation [6] by $3(p_1' - 5)$, we have

$$\left[3(p_1' - 5)\right]p_1' + 15p_1' + 10(3p_1' - 10) + 2(3p_1' - 10)\left[3(p_1' - 5)\right] = 67.5\left[3(p_1' - 5)\right]$$

$$3p_1'^2 \cancel{-15p_1' + 15p_1'} + 30p_1' - 100 + 6(3p_1'^2 - 25p_1' + 50) = 202.5p_1' - 1012.5$$

$$3p_1'^2 + 30p_1' - 100 + 18p_1'^2 - 150p_1' + 300 - 202.5p_1' + 1012.5 = 0$$

This reduces to the quadratic equation

$$21p_1'^2 - 322.5p_1' + 1\,212.5 = 0$$

which has solutions $p_1' = 8.784$ cm and 6.573 cm.

Case 1: $p_1' = 8.784$ cm

$\therefore\ p_1' - p_1 = 8.784\text{ cm} - 7.50\text{ cm} = 1.28\text{ cm}$

From [4]: $q_2' = 32.7$ cm

$\therefore\ q_2' - q_2 = 32.7\text{ cm} - 15.0\text{ cm} = 17.7\text{ cm}$

Case 2: $p_1' = 6.573$ cm

$\therefore\ p_1' - p_1 = 6.573\text{ cm} - 7.50\text{ cm} = -0.927\text{ cm}$

From [4]: $q_2' = 19.44$ cm

$\therefore\ q_2' = q_2 = 19.44\text{ cm} - 15.0\text{ cm} = 4.44\text{ cm}$

From these results it is concluded that:

> The lenses can be displaced in two ways. The first lens can be moved 1.28 cm farther from the object and the second lens 17.7 cm toward the object. Alternatively, the first lens can be moved 0.927 cm toward the object and the second lens 4.44 cm toward the object.

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P36.95 (a) The lens makers' equation, $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$, becomes:

$$\frac{1}{5.00\text{ cm}} = (n-1)\left[\frac{1}{9.00\text{ cm}} - \frac{1}{(-11.0\text{ cm})}\right]$$

giving $n = \boxed{1.99}$.

(b) As the light passes through the lens for the first time, the thin lens equation, $\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{f}$, becomes:

$$\frac{1}{8.00\text{ cm}} + \frac{1}{q_1} = \frac{1}{5.00\text{ cm}}$$

giving $q_1 = 13.3\text{ cm}$, and $M_1 = -\frac{q_1}{p_1} = -\frac{13.3\text{ cm}}{8.00\text{ cm}} = -1.67.$

This image becomes the object for the concave mirror with:

$$p_M = 20.0\text{ cm} - q_1 = 20.0\text{ cm} - 13.3\text{ cm} = 6.67\text{ cm}$$

and $f = \frac{R}{2} = +4.00\text{ cm}.$

The mirror equation becomes: $\frac{1}{6.67\text{ cm}} + \frac{1}{q_M} = \frac{1}{4.00\text{ cm}}$,

giving $q_M = 10.0\text{ cm}$,

and $M_2 = -\frac{q_M}{p_M} = -\frac{10.0\text{ cm}}{6.67\text{ cm}} = -1.50.$

The image formed by the mirror serves as a real object for the lens on the second pass of the light through the lens, with

$$p_3 = 20.0\text{ cm} - q_M = +10.0\text{ cm}$$

The thin lens equation yields: $\frac{1}{10.0\text{ cm}} + \frac{1}{q_3} = \frac{1}{5.00\text{ cm}}$,

or $q_3 = 10.0\text{ cm}$

and $M_3 = -\frac{q_3}{p_3} = -\frac{10.0\text{ cm}}{10.0\text{ cm}} = -1.00.$

The final image is a real image located

$\boxed{10.0\text{ cm to the left of the lens}}$.

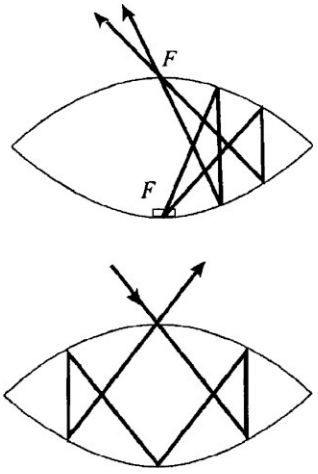
(c) From above, we find the overall magnification:

$$M_{total} = M_1 M_2 M_3 = \boxed{-2.50}$$

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(d) The overall magnification is negative, so the final image is inverted.

P36.96 (a) The object is located at the focal point of the upper mirror. Thus, the upper mirror creates an image at infinity (i.e., parallel rays leave this mirror). For the upper mirror, the object is real, and the mirror equation, $\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$, gives



ANS. FIG. P36.96

$$\frac{1}{7.50\text{ cm}}+\frac{1}{q_1}=\frac{1}{7.50\text{ cm}}$$

$$\rightarrow\ q_1 \approx \infty \text{ (very large)}$$

The lower mirror focuses these parallel rays at its focal point, located at the hole in the upper mirror. For the lower mirror, the object is virtual (behind the mirror), $p_2 \approx -\infty$:

$$\frac{1}{-\infty}+\frac{1}{q_2}=\frac{1}{7.50\text{ cm}}\ \rightarrow\ q_2 = 7.50\text{ cm}$$

The overall magnification is

$$M = m_1 m_2 = \left(\frac{-q_1}{p_1}\right)\left(\frac{-q_2}{p_2}\right)=\left(\frac{\infty}{7.50\text{ cm}}\right)\left(\frac{7.50\text{ cm}}{-\infty}\right)=-1$$

Thus, the image is real, inverted, and actual size.

(b) Light travels the same path regardless of direction, so light shined on the image is directed to the actual object inside, and the light then reflects and is directed back to the outside. Light directed into the hole in the upper mirror reflects as shown in the lower figure, to behave as if it were reflecting from the image.

P36.97 First, we solve for the image formed by light traveling to the left through the lens. The object distance is $p_L = p$, so

$$\frac{1}{p_L}+\frac{1}{q_L}=\frac{1}{f_L}\ \rightarrow\ \frac{1}{q_L}=\frac{1}{f_L}-\frac{1}{p}$$

Next, we solve for the image formed by light traveling to the right and reflecting off the mirror. The object distance is $p_M = d - p$, so

$$\frac{1}{p_M}+\frac{1}{q_M}=\frac{1}{f_M}\ \rightarrow\ \frac{1}{q_M}=\frac{1}{f_M}-\frac{1}{p_M}=\frac{p_M - f_M}{f_M p_M}$$

$$q_M=\frac{f_M p_M}{p_M - f_M}=\frac{f_M(d-p)}{d-p-f_M}$$

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If q_M is positive (real image), the image formed by the mirror will be to its left, and if q_M is negative (virtual image), the image formed by the mirror will be to its right; for either case, the image formed by the mirror acts as an object for the lens at a distance p'_L:

$$p'_L = d - q_M = d - \frac{f_M(d-p)}{(d-p)-f_M} = \frac{d(d-p-f_M)-f_M(d-p)}{d-p-f_M}$$

We solve for the position of the final image q'_L:

$$\frac{1}{q'_L} = \frac{1}{f_L} - \frac{1}{p'_L} = \frac{1}{f_L} - \frac{d-p-f_M}{d(d-p-f_M)-f_M(d-p)}$$

For the two images formed by the lens to be at the same place,

$$\frac{1}{q_L} = \frac{1}{q'_L} \rightarrow \frac{1}{f_L} - \frac{1}{p_L} = \frac{1}{f_L} - \frac{1}{p'_L} \rightarrow p'_L = p_L$$

Therefore,

$$\frac{d(d-p-f_M)-f_M(d-p)}{d-p-f_M} = p$$

$$d(d-p-f_M)-f_M(d-p) = p(d-p-f_M)$$

$$d^2 - pd - f_M d - f_M d + f_M p = pd - p^2 - f_M p$$

$$d^2 - 2(p+f_M)d + (2f_M p + p^2) = 0$$

Solving for d then gives

$$d = \frac{2(p+f_M) \pm \sqrt{4(p+f_M)^2 - 4(1)(2f_M p + p^2)}}{2(1)}$$

$$d = \frac{2(p+f_M) \pm \sqrt{4p^2 + 8f_M p + 4f_M^2 - 8f_M p - 4p^2}}{2}$$

$$d = \frac{2(p+f_M) \pm \sqrt{4f_M^2}}{2} = (p+f_M) \pm f_M$$

Therefore, $\boxed{d = p \text{ and } d = p + 2f_M}$.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P36.2 4.58 m

P36.4 (1) 4.00 m; (2) 12.00 m; (3) 16.00 m

P36.6 See ANS. FIG. P36.6 for the locations of the five images.

P36.8 (a) 33.3 cm in front of the mirror; (b) –0.666; (c) real; (d) inverted

P36.10 (a) See ANS FIG P36.10; (b) q = –40.0 cm, so the image is behind the mirror; (c) M = +2.00, so the image is enlarged and upright; (d) See P36.10(d) for full explanation.

P36.12 (a) –26.7 cm; (b) upright; (c) 0.026 7

P36.14 (a) +2.22 cm; (b) +10.0

P36.16 A convex mirror *diverges* light rays incident upon it, so the mirror in this problem cannot focus the Sun's rays to a point.

P36.18 (a) 0.708 m in front of the sphere; (b) upright

P36.20 (a) $\frac{ad}{a^2-1}$; (b) $\frac{ad}{a^2-1}$

P36.22 (a) 8.00 cm; (b) See ANS. FIG. P36.22(b); (c) virtual

P36.24 (a) 16.0 cm from the mirror; (b) +0.333; (c) upright

P36.26 (a) See P36.26(a) for full explanation; (b) real image at 0.639 s and virtual image at 0.782 s

P36.28 8.05 cm

P36.30 38.2 cm below the top surface

P36.32 3.75 mm

P36.34 See P36.34 for full explanation.

P36.36 (a) (i) 3.77 cm from the front of the wall, in the water, (ii) 19.3 cm from the front wall, in the water; (b) (i) +1.01, (ii) +1.03; (c) The plastic has uniform thickness, so the surfaces of entry and exit for any particular ray are very nearly parallel. The ray is slightly displaced, but it would not be changed in direction by going through the plastic wall with air on both sides. Only the difference between the air and water is responsible for the refraction of the light; (d) yes; (e) If $p = |R|$, then $q = -p = -|R|$; if $p > |R|$, then $|q| > |R|$. For example, if $p = 2|R|$, then $q = -3.00|R|$ and M = +2.00.

P36.38 (a) 650 cm, real, inverted, enlarged; (b) –600 cm, virtual, upright, enlarged

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P36.40 (a) 12.3 cm to the left of the lens; (b) 0.615; (c) See ANS. FIG. P36.40.

P36.42 (a) The image is in back of the lens at a distance of 1.25f from the lens; (b) –0.250; (c) real

P36.44 (i) See ANS. FIG P36.44(i): (a) 20.0 cm in back of the lens, (b) real, (c) inverted, (d) $M = -1.00$, (e) Algebraic answers agree, and we can express values to three significant figures: $q = 20.0$ cm, $M = -1.00$; (ii) See ANS. FIG. P36.44(ii): (a) 10 cm front of the lens, (b) virtual, (c) upright, (d) $M = +2.00$, (e) Algebraic answers agree, and we can express values to three significant figures: $q = -10.0$ cm, $M = +2.00$, (f) Small variations from the correct directions of rays can lead to significant errors in the intersection point of the rays. These variations may lead to the three principal rays not intersecting at a single point.

P36.46 (i): (a) 13.3 cm in front of the lens, (b) virtual, (c) upright, (d) +0.333; (ii): (a) 10.0 cm in front of the lens, (b) virtual, (c) upright, (d) +0.500; (iii): (a) 6.67 cm in front of the lens, (b) virtual, (c) upright, (d) +0.667

P36.48 $dq = -\dfrac{q^2}{p^2}dp$

P36.50 (a) $q_a = 26.3$ cm, $q_d = 46.7$ cm, –8.75 cm, –23.3 cm; (b) See ANS. FIG. P36.50(b); (c) See P36.50(c) for full explanation; (d) The integral stated adds up the areas of ribbons covering the whole image, each with vertical dimension $|h'|$ and horizontal width dq; (e) 328 cm^2.

P36.52 See P36.52 for full explanation.

P36.54 (a) –34.7 cm; (b) –36.1 cm

P36.56 $f/1.4$

P36.58 (a) –4.00 diopters; (b) diverging lens

P36.60 (a) –25.0 cm; (b) nearsighted; (c) –3.70 diopters

P36.62 (a) $+50.8 \text{ diopters} \le P \le 60.0$ diopters; (b) –0.800 diopters, diverging

P36.64 The image is inverted, real, and diminished.

P36.66 (a) 4.17 cm; (b) 6.00

P36.68 (a) –800; (b) inverted

P36.70 (a) See P36.70(a) for full explanation; (b) $-\dfrac{hf}{p}$; (c) –1.07 mm

P36.72 –25.0 cm

P36.74 $f = \dfrac{-Md}{(1-M)^2}$ when the lens is diverging; $f = \dfrac{Md}{(M-1)^2}$ when the lens is converging

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P36.76 (a) 0.833 mm; (b) 0.820 mm

P36.78 (a) 13.3 cm in front of the first lens; (b) –6.00; (c) inverted; (d) virtual

P36.80 (a) $x' = \dfrac{1024 - 58.0x}{6.0 - x}$ where x and x' is are in centimeters; (b) See P36.80(b) for full explanation; (c) The image moves to infinity and beyond—meaning it moves forward to infinity (on the right), jumps back to minus infinity (on the left), and then proceeds forward again; (d) The image usually travels to the right, except when it jumps from plus infinity (right) to minus infinity (left).

P36.82 (a) See P36.82(a) for full explanation; (b) 1.75 times

P36.84 q = 10.7 cm

P36.86 See P36.86 for full explanation

P36.88 $\dfrac{p(f_1 + f_2) - f_1 f_2}{p - f_1}$

P36.90 (a) $\dfrac{1}{f} = \dfrac{1}{p_1} + \dfrac{1}{1.50 - p_1}$; (b) $\dfrac{1}{f} = \dfrac{1}{p_1 + 0.900} + \dfrac{1}{0.600 - p_1}$; (c) 0.300 m; (d) 0.240 m

P36.92 (a) 20.0 cm; (b) –6.00; (c) inverted; (d) q_2 = 6.67 cm and M_{overall} = –2.00, inverted

P36.94 (a) 67.5 cm; (b) The lenses can be displaced in two ways. The first lens can be moved 1.28 cm farther from the object and the second lens 17.7 cm toward the object. Alternatively, the first lens can be moved 0.927 cm toward the object and the second lens 4.44 cm toward the object.

P36.96 (a) The image is real, inverted, and actual size; (b) Light travels the same path regardless of direction, so light shined on the image is directed to the actual object inside, and the light then reflects and is directed back to the outside. Light directed into the hole in the upper mirror reflects as shown in the lower figure, to behave as if it were reflecting from the image.

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